

AMERICAN BEE JOURNAL.

EDITED AND PUBLISHED BY SAMUEL WAGNER, WASHINGTON, D. C.

AT TWO DOLLARS PER ANNUM, PAYABLE IN ADVANCE.

VOL. VI.

NOVEMBER, 1870.

No. 5.

AMERICAN BEE JOURNAL.

EDITED AND PUBLISHED BY SAMUEL WAGNER, WASHINGTON, D. C.

AT TWO DOLLARS PER ANNUM, PAYABLE IN ADVANCE.

VOL. VI.

NOVEMBER, 1870.

No. 5.

The Project Gutenberg eBook of The American Bee Journal, Vol. VI., Number 5, November 1870

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

Title: The American Bee Journal, Vol. VI., Number 5, November 1870

Author: Various

Editor: Samuel Wagner

Release date: October 29, 2018 [eBook #58189]

Language: English

Other information and formats: www.gutenberg.org/ebooks/58189

Credits: Produced by Wayne Hammond and The Online Distributed Proofreading Team at <http://www.pgdp.net> (This file was produced from images generously made available by The Internet Archive)

*** START OF THE PROJECT GUTENBERG EBOOK THE
AMERICAN BEE JOURNAL, VOL. VI., NUMBER 5, NOVEMBER 1870

AMERICAN BEE JOURNAL.

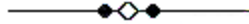
EDITED AND PUBLISHED BY SAMUEL WAGNER,
WASHINGTON, D. C.

AT TWO DOLLARS PER ANNUM, PAYABLE IN ADVANCE.

VOL. VI.

NOVEMBER, 1870.

No. 5.



[For the American Bee Journal.]

Cure of Foulbrood.

Mr. EDITOR:—I promised, (vol. V., page 187,) to report how my refrigerator wintered its colony. The frames were covered with a piece of old carpeting, and the whole space outside the inner hive packed with straw and shavings. This spring it was in splendid condition, and it was found necessary to remove brood and cut out queen cells as early as the 20th of May; and, for this locality, the surplus would have been large, if I had not been obliged to break up the colony on account of *foulbrood*.

You can imagine my disappointment when my apiarian friend, Mr. Sweet of West Mansfield, pointed out to me this loathsome disease in my choicest Italian colony, early in June, when up to that time I had supposed that everything was prosperous with my twelve colonies. After a thorough examination I found six hives more or less affected, and according to high authority, should be condemned to death. The other six appeared free from disease at this time, although three more subsequently became diseased.

This is my second summer of bee-keeping, and all the duties pertaining to an apiary were entered into with the enthusiasm, and shall I confess it, the ignorance and carelessness of a novice. Yes, ignorance and culpable carelessness, for in gathering empty combs from various quarters, the disease was introduced and spread among my pets. One hive, in particular, of empty comb had the peculiar odor, perforated cells, and brown viscid fluid, with which I have since become so familiar this summer; and it seems unaccountable to me, how any person with the Bee Journal wide open and Quinby's instructions before him, could be so careless as to give such combs to his bees.

But such was the fact, and foulbrood spreading right and left. What shall be done to get rid of it? Shall Quinby be followed, purify the hive and honey by scalding, and treat the colony as a new swarm; or shall the heroic treatment of Alley be adopted; bury or burn bees and hive, combs and all? The latter has sent me some fine queens; but the former has always given

reliable advice, and I shall follow his instructions with two colonies which are past all cure, and reserve the others for treatment, hoping that I may find some cure, or at least palliative for the disease, and add my mite of experience, and, perhaps, useful knowledge to our Bee Journal.

Accordingly, June 8th, the combs of the two condemned colonies were melted into wax, the honey drained over and scalded, and the bees, after a confinement of forty hours, were treated like new swarms; and now, September 18th, are perfectly healthy and in fine condition for winter.

I will not occupy your valuable space with all the details of my experiments and fights (which lasted through three months) with the trials of doses of different strengths and kinds, with old comb and new, with young queens and old ones, and with no queen at all, and how, in doing this, I was obliged to keep up the strength of the colony for fear of robbers and of spreading the disease to my neighbors. Suffice it to say, that after two months I had made no apparent headway, although still determined to “fight it out on this line, if it took all summer” and my last hive. In fact, I devoted my apiary to the study of this disease, and, perhaps, death.

Starting with, and holding to the theory that foulbrood is contagious only by the diffusion of living germs of feeble vitality, (and I was strengthened in my conjecture in microscopical examinations, by finding the dead larvæ filled with nucleated cells,) I determined to try those remedies which have the power of destroying the vitality of these destructive germs, these living organisms. And no remedies seemed to me more potent than carbolic acid and hyposulphite of soda. At first I used both, making one application of each, with an interval of one day, and with apparent benefit. But, attributing the improvement to the more powerful of the two, I abandoned the hyposulphite and used the carbolic acid alone, and I was so infatuated with the idea of its superiority, that I did not give it up until three of the four hives had become so hopelessly diseased, that the combs were destroyed and the colonies treated to new combs (as it was late in the season,) and freely fed with sugar and water. These are now in good condition for winter.

The fourth hive was carried a mile away, the queen caged, and the colony strengthened with a medium sized second swarm. After all the brood, which was advanced, had left the cells, I transferred the colony to a clean hive;

thoroughly sulphured the old hive with burning sulphur, and stored it away in a safe place for future experiments. I now thought my apiary free from the pest; but on thoroughly examining the whole, three new cases of foulbrood were found—one very badly affected, and two slightly so, with perhaps twenty to forty cells diseased and perforated.

Entered according to Act of Congress, in the year 1870, by Samuel Wagner, in the office of the Librarian of Congress, at Washington.

This was about the 1st of August, and again hyposulphite of soda was selected for the trial; and from the first application I have had the disease under control. Three days ago I examined the three colonies thoroughly, and found no new cells diseased in the two which had been the least affected; and in the almost hopelessly diseased one (as much diseased, in fact, as any of those that I destroyed,) an entire brood had been raised, with not over fifty or sixty diseased and perforated cells with dead larvæ remaining, most on one comb, and nearly all the cells contained a new supply of eggs; this colony is certainly convalescent, and I think now, from the recent and second application of the hyposulphite of soda, is entirely cured. Still, I should not be surprised to find two or three, or even more, perforated cells after this second crop of brood has hatched, as the whole hive, honey, and comb, had been for so long a time so thoroughly saturated with the disease, and at least two-thirds of the cells had, before the *medicine* was used, been filled with putrid larvæ. If so, I shall treat it to a third dose.

Now, Mr. Editor, as it is frequently of as much practical importance to tell how to administer a remedy, as it is to know its name, I will ask your indulgence a little longer, hoping that others may improve upon my remedy or at least test it, if they are so unfortunately ignorant and careless as I was, in bringing “the wolf home to the fold.”

The solution of hyposulphite of soda which I used, was one ounce to half a pint of rain water. With this I thoroughly washed out every diseased cell with an atomizer, after opening the cap; also spraying over the whole of the combs and the inside of the hive. The instrument I use is a spray producer, invented by Dr. Bigelow of Boston, and sold by Codman & Shurtleff of that city. There are two small metallic tubes, a few inches long, soldered together; and by placing the point of exit of the spray at the lower part of

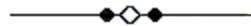
the cell, the whole of the contents of the cell is instantly blown out upon the metallic tubes. With a very little practice there is no necessity for polluting the comb with the putrid matter. Place the comb perfectly upright or a little leaned towards you, and there is no difficulty; yet, if a drop should happen to run down the comb, it would do no harm, but had better be carefully absorbed with a piece of old dry cotton cloth. I quite frequently do this with the bees on the comb, as it does them no harm, to say the least, to get well covered with the vapor.

It is not at all injurious to the larvæ, after they are two or three days old, though it may be before that time, as I have noticed that after using the hyposulphite where there are eggs and very young larvæ, the next day the cells are perfectly clean.

There are many interesting points which have come up during my summer's fight, which I would speak of; but I have already gone beyond all reasonable bounds in this communication.

EDWARD P. ABBE.

New Bedford, Mass., Sept. 18, 1870.



[Translated from the *Bienenzeitung*,
For the American Bee Journal.]

Queen Breeding.

To obtain not only purely fertilized queens, but fine, bright yellow ones, I have for some years proceeded thus:

As all Italian queens do not produce equally fine drones, I mark those stocks in the course of the summer which contain queens producing the choicest of these. Then, in the following spring, when I desire to have a plentiful supply of prime Italian drones early, and before common drones make their appearance in neighboring apiaries, insert in the hives thus selected and marked, combs of worker brood taken from other colonies. I do this in order to make those colonies very populous, so as to induce drone-egg-laying; for a queen will always be disposed to commence doing so, if she is in a strong colony well supplied with honey, or is well fed. As soon as I find that those colonies are becoming populous under this management, I insert some empty drone comb in the centre of the brooding space. These the queen, stimulated by liberal feeding, will speedily supply with eggs; and when the drone brood so produced is nearly mature, I subdivide these combs and insert pieces in nuclei previously furnished with young bees, worker brood, and eggs, taken from the colonies containing the choice queens from which I design to breed, and which are known to produce the largest, most active, and best marked workers.

As the drones form the brood thus introduced mature several days sooner, than the young queens bred in the same nuclei, there is a strong probability that the latter will be fertilized by them and consequently produce fully marked choice progeny, as it is certain that queens will almost invariably be fertilized early if they and the drones are bred in the same hive or nucleus, since that secures the simultaneous flight of both and obviates the necessity of a wide range in their excursions. I adopt this process also, because if the Italian drones of the colonies, which contain the young queens, are poorly marked and dark yellow in color, we cannot reasonably look for bright and handsomely marked progeny.

At about ten o'clock in the morning of a calm, clear day, when the young queen is at least two days old, I feed the bees of the nucleus with diluted honey. Drones and queens will then almost invariably issue at the same time, and before common drones from other colonies or neighboring apiaries are on the wing. Thus both disappointment and delay are in a great measure precluded. I do not stimulate the bees of the nucleus by feeding either on the first or the second day after a young queen has left her cell, because she is then yet too feeble to make an excursion with safety. But I have frequently succeeded in having fertilization effected on the third or fourth day, in favorable weather, when the nucleus thus stimulated contained both drones and queen; and in many cases the queens began to lay on the third or fourth day thereafter. In this way, I not only obtain many (I do not say all) purely fertilized queens; but also very superior ones, large, vigorous, and prolific, producing both workers and drones well marked and brightly colored.

I do not indeed claim that this process gives us absolute certainty, but only a very great probability, that the queens we rear will be purely fertilized. Other bee-keepers too, who employed it long before the Kœhler method was promulgated, regard it as furnishing the most likely means of assuring success. Thus, for instance, the President of the Bee-keeper's Union of Moravia, Dr. Ziwanski, who is not a blind imitator of others, but a careful and indefatigable inquirer, never recommending aught for adoption till he has himself tested it with success, found my method worthy of adoption five years ago already, for his annual report for 1865 contains the following passage:—

“I made five nuclei this year, with fresh brood from pure original Italians. When fitting them up, I recollected a suggestion of the Rev. Mr. Stahala, and inserted both drone and worker brood in four of them, omitting the drone brood in the fifth. The queens of the first four mentioned were purely fertilized, while the one in the fifth nucleus mated with a common drone. This result induces me to invite your attention to the fact, for it is reasonable to presume that queens making their excursions will be more likely to mate with drones from their own hives flying simultaneously, than with drones from other and distant hives. The queen usually makes such excursions only at periods when drones are flying, and there is then

generally great commotion in the hive, as though there was much eagerness to get abroad and enjoy the genial air. Still, too much must not be expected from this suggestion and its adoption. It is not supposed that any preliminary arrangements or appointments are made by drones or queens, before the excursion is undertaken; but merely that there is a much greater probability that parties flying at the same time and necessarily in close proximity, will mate, than those starting from remoter points. Hence since it can do no possible harm to supply our nuclei with drone and drone-brood in this manner, the plan should by no means be disregarded when preparing to Italianize an apiary.”

By means of this process, having selection to a great degree in my power, I frequently obtain queens nearly entirely yellow, having black only at the extremity of the abdomen. I have procured queens for breeding from both Dzierzon and Mona. The young queens breed from Dzierzon’s stock were at first handsomer than those bred from Mona’s. But in later years, since using the method I now recommend, I obtain equally fine queens from the latter’s stock. The drones from Mona’s queens were, from the start yellower than those from Dzierzon’s, which were only faintly tinged with yellow on the sides, and had dark orange bands. Observing this, I then took worker brood and queen cells from the Dzierzon’s queens, with drones and drone-brood from the Mona queens, to furnish the same nucleus, and thus obtained regularly very handsome queens, bright workers, and very fine drones.

J. STAHALA, *Pastor*.

Dolein, near Olmutz, Feb. 5., 1870.



[For the American Bee Journal.]

Purity of Italian Queens.

Your correspondent, E. L. Briggs, in the August number of the Journal, has stirred up the bee-keepers a little; and for fear they will not discuss the point which most interests me, I drop you a line, hoping that those who have had more experience may be able to settle the question.

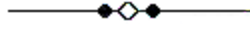
It is a fact which I think no one will deny, that it would be for the interest of every one selling queens, to send only such as are purely fertilized. It being as easy to rear queens from pure eggs as from any other, we may look to some other cause than selfishness or cheapness of the price for the difficulty. I have managed my apiary under the impression that the Dzierzon theory is correct, that the drones from a pure queen which had mated with a black drone, were pure.

I have failed in keeping my stock pure enough to breed from; and in my opinion, other bee-keepers who have reared queens in the same way, are as badly off as myself. If we wish to improve the Italian bee, we may do so by selecting the best of its race, both male and female, to breed from; not by crossing with the black bee. The type of the Italian bee should be so fixed, that the bees all show the same marking. We may fix the type of any admixture of the German and Italian bees, so that they will have similar markings. The crossing has been so recent in many cases, that there is no uniformity of color. Breeders of choice stock look as much to the quality and purity of the male as the female parent. It is my present belief that bees are as much subject to the rule, as the animal creation are.

I look for higher results than any yet attained, when we control (as we soon shall) the mating of our queens; and the low priced ones have given me the most satisfaction so far.

L. C. WHITING.

East Saginaw, Mich.



[For the American Bee Journal.]

Italian Queens.

MR. EDITOR:—Since so much has been said of late about Italian queens, (especially cheap ones,) I feel it my duty, in justice to Mr. Alley, to say, that I purchased one of his \$2.50 queens last June and have bred sixteen queens from her, besides a host of drones and workers; and the facts are, first, her progeny are all three-banded; second, she is the most prolific queen in my apiary; third, her workers are very industrious; fourth and last, I am not at all out of patience because she cost me only \$2.50. Five dollars will not buy her to-day; and if I have the good luck to keep her till next June (supposing she is young, as claimed by Mr. Alley), I shall not want to part with her for two fives. All who have seen her and her workers, pronounce them beauties; and Italian bees are nothing new in these parts.

JAMES HEDDON.

Dowagiac, Mich., Sept., 1870.



[For the American Bee Journal.]

Novice.

Mr. EDITOR:—Sometime ago, in one of our articles, we mentioned that we considered the “Apiary” department in the “*Rural New Yorker*” of more real worth than some of the periodicals specially devoted to bees.

We had then seen about half a dozen of the “Rurals” that contained some very good articles, from the pens of intelligent bee-keepers who were well up to the times. Since then, however, we have seen so much else there so greatly behind the times, that we must think our decision then a little hasty. For instance last week a bee-keeper takes the trouble to inform the public that “hives should be moved in the night *when the bees are all in*, for he had just moved some in the day time and a large number that were out, never found their hive on their return. So take notice everybody, always move your bees at night!” As this was given as a piece of valuable information, we looked in vain for some note from the editors, cautioning their readers against falling into the same error, and pointing it out. And then we wondered if the editors knew any better, or anything about bees at all, for many of their articles seem to imply that they are uninformed and publish anything they come across, indiscriminately, truth and error, without note or comment.

The editor of the *Apiculturist* thought it the height of absurdity because we seemed to consider him in any way responsible for what his correspondent wrote. We certainly *were* so innocent as to suppose that an editor *knew* what he was going to publish, and that should a correspondent send him an article containing a very gross error, calculated to lead beginners astray, he would tell such correspondent his mistake, without using his article; or if it contained something else good and valuable, and he decided to publish it, he would kindly mention the mistake or error, in a little note somewhere, and give his readers confidence by letting them know that some one *was* “running the machine” “somewhere.”

There are a large number of good farmers who refuse to read agricultural papers, because they say, and with considerable reason, that more than half that is written is “impracticable nonsense.” We believe the American Agriculturist and the American Bee Journal are at least two noble exceptions. None of their readers can fail to know that each of those papers is *edited* by some one who is fully posted, and is *at home* too *every time*.

The Apiculturist intimates that we think no one else has a right to *start* a bee journal. So far from that we would be glad to subscribe this minute for half a dozen more; if they were in charge of competent men and had the broad platform before them that our own Journal has—namely, the advancement of bee culture for the nation at large.

We should have replied to the Apiculturist before, but he “called names,” and when we were a small boy we used to make it a principle that when our comrades called us names, we “wouldn’t play any more,” and we feel just so still.

We, too, Mr. Editor, noticed the mention in the “Scientific American,” of the chicken roost bee arrangement to stop moths, and felt pained to think that anything, so far behind the times, should be found in that paper. Then, again, we noticed shortly after where they advised a correspondent to *chop* up his combs and strain the honey out, and mentioned too that it was *said* that the outside combs contained the nicest honey! Have Munn & Co., too, been sleeping in Rip Van Winkle style, or do they think us Bee Journal people not to be depended on?

We have had many letters from highly intelligent people, even professors in colleges, asking about the melextractor and inquiring whether there was no serious objection to such unnatural treatment of bees?

“Unnatural treatment,” indeed! About the 25th of last June, a farmer called on us to know where he could sell his honey best. On asking him how he had got it so early, he coolly informed us that he had *taken it up*, as it *seemed full*! But how about the brood? He didn’t know what we meant by brood, but had thrown away the young bees and did not think that they were of any use! Murdered thousands of young innocents before the end of June!

Of course such treatment is perfectly *natural* and right. He didn't get much for his honey.

Mr. Editor, we are getting hoarse in trying to explain, and all we tell inquirers now is to get the "*American Bee Journal*." Yet many, many times they can't afford it, and many more times don't get time to read it. Yet the same persons will say—"Why, Novice, your forty-six hives of bees have been worth more to you than any hundred acre farm in Medina county," and go home quite excited.

We have had a few weeks' drouth, the first this season, and it soon stopped the honey from autumnal wild flowers.

Since Mr. Tillinghast suggested our being called "Expert" (or some such foolishness), we think we could hardly be honest without confessing some of our work this fall. For instance, we removed queen from No. 23, August 9th, and ten days after cut out thirty-two (32) queen cells. We have mentioned before that we tried hatching some of them in cages, and the rest were put in hives from which we had removed hybrid queens. We were such an *expert* at the business that we hatched about one-half the thirty-two, and after they were hatched, we *bungled* the life out of *every one*—some by artificial fertilization experiments; and the rest wouldn't lay and finally died their "own selves."

Well, (we have considerable patience,) we tried again; removed queen from No. 16, August 28, and cut out twenty-one (21) cells ten days after. Of these we *did* raise five laying queens; and most of the other cells were destroyed by laying them on the top of the frames when the weather was too cool. In fact we have had more cells destroyed this fall than ever before, and only saved five by inserting them carefully in place of one *cut out*. Now, Mr. Editor, we should have felt somewhat better at this result, had we not discovered that the original queen removed from No. 16 had been killed, and only a miserable, small, black queen reared in her place. She was put in a hive in which we had a caged, unfertile queen, and we neglected to look whether they had raised any more. *Inexcusable carelessness*, we call it.

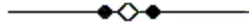
To shorten the matter, we sent Mr. Grimm fifty dollars on Monday morning, and received twenty-five nice queens (or a part of them at least) on

Saturday afternoon. Is not that pretty prompt?

Now, Mr. Editor, we are going to take this queen raising business up next spring just where we left off; and if we can't do better, and at least raise enough for our own apiary, we shall call ourself something worse than

NOVICE.

October 10, 1870.



[For the American Bee Journal.]

Natural, prolific, and hardy Queens.

PART 3.

Answer to Charles Dadant and Willard J. Davis, in September number of the American Bee Journal, pages 60 and 61.

To commence with Mr. Dadant. He says, first, that “we are all disposed to regard our own ideas as indisputable.”

Answer. Prove all things; then hold fast to the true. Do not condemn before trial. I have been several years experimenting and am satisfied with my method, as a means of procuring natural, prolific, hardy and long-lived queens—far, far ahead of any yet given to the public. It having relieved me from the disappointment and losses heretofore experienced in artificial swarming, with forced or artificial queens, I have freely given my mode to the public, for adoption or rejection, as they see fit. Those who are *set* in their way, are under no obligation to either adopt or even try my mode; but there are those who are not satisfied with their present light, and who will be benefited by the knowledge of an improved process, and to them my communications are addressed.

He says, second, that I “condemn all artificially raised queens.”

Answer. I do: as against nature, reason, and common sense. I see a difference in a provision of nature, by means of which a swarm, accidentally deprived of its queen, can temporarily replace her, till one can be raised in a more natural way, and the way men in their *wisdom* are running the race out. You yourself prove my position by almost every line of your article, if you would only place your trials, troubles, vexations, and losses to their right account—*forced or artificially raised queens*. New brood may seemingly save you for a time; but when all breeders have the *cholorosis* stamped on the product of their apiaries, like will beget like.

He says, in the third place—“why does friend Price imagine that artificial queens are not as good as natural ones?”

Answer. Because convinced by years of experiment and careful comparison (not hard to see, I assure you) of natural with forced queens raised by the means you have mentioned in your article, and by others not mentioned. Even now I am trying the experiment of raising forced queens from the brood of a pure Italian queen received last spring from a celebrated breeder. But so far I have only succeeded in raising cripples, drone layers, and non-egg-hatching queens. Most of them *play out* before commencing to lay; yet I have raised them from the egg—not one of them hatching before the sixteenth day.

He says, fourth, after giving away or getting queens from the egg, “I guess this method is as good as, and more simple than, that of friend Price.”

Answer. You would go through every motion that I do, and get two or three queens, worthless in comparison with natural ones; while I would secure from ten to sixty natural ones. If you followed your own method, you would have to divide almost every hive in your apiary, if you got through swarming in any season; while by my method¹ one hive would furnish all the natural queen cells that would be wanted in the largest apiary in the time of natural swarming.

He says, fifth, “a queen hatched from grubs three or four days old is just as good as any.”

Answer. To sell!

Sixth, he says, “many bee-keepers find the half-blood Italian bees are better than pure ones”—his reason being that in and in breeding is broken up.

Answer. Those that receive them, let them swarm naturally; thus the forcing is at an end, and nature again asserts her superiority.

He says, seventh, “In good seasons the queens raised in small nuclei are as good as those raised in full stocks.”

Answer. He admits that they cannot at all times raise good ones. He had better have attributed it to the lack of a natural instinct to raise good ones. A swarm on the eve of swarming, broken up into nuclei, would probably raise pretty fair queens—say half as good as natural ones. As well might you hire a rough wood chopper or ditcher to make a watch, as to set a nucleus of bees not having the swarming instinct, to raise a first rate chronometer balanced queen.

Mr. W. J. Davis says that he does not know what effect my Revolvable, Reversible, Double-cased, Sectional Bee-hive may have had on the tender life of a young queen, *forced or artificial*.

As I have only used my old Langstroth hives for nuclei; *my* hive has of course not had any influence on them, for good or evil. But my twenty young natural queens, raised by my method, are without exception hardy, prolific, and have every promise of being long-lived. Had they been forced queens two-thirds of them would have been played out before this time. They are as prolific as any of my old “natural” queens which I bought of those who practice natural swarming only. My R. R. D. C. S. B. Hive has a good effect on the life of natural queens; and as Mr. Dadant says his bees in my hive have done better than in any other, and he has of several patents, and as he says he has only raised forced queens, my R. R. D. C. S. Bee-hive most probably saved him.

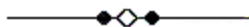
Secondly, after reading all his conditions of age, weather, season, stock, nuclei, time, and egg, that have to be consulted to insure a good queen by the forcing process, I have an idea that his queens are natural ones. Do you not bring your bees up to swarming and then secure their cells Gallup fashion? Gallup calls such natural queens. I should. Otherwise why not have good queens from March to October?

Thirdly, Mr. Davis says that “if Mr. Price *or any other* man will, upon examination, decide correctly, by size or fertility (amount of brood), which are of the former and which are of the latter class, he may pick out ten as large and yellow queens as he *ever saw*, and I will make him a present of the same.”

Answer. I have only one artificial queen laying, my pure *prolific* Italian. I will guarantee any of my black, “young or old,” or other natural queens, to fill five frames with brood quicker than she can fill one; and if you, or “any other man,” cannot see any difference between my forced queens² and my natural ones, you must be deficient in the organs of size and weight, and would not be able to tell a Shetland pony from an elephant.

JOHN M. PRICE.

Buffalo Grove, Iowa.



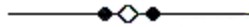
[For the American Bee Journal.]

Introducing Queens.

Dr. H. C. Barnard in the June number of the A. B. Journal, gave directions for introducing queens by fumigating with tobacco smoke. I had introduced them by means of the queen cage, and sprinkling them with sweetened water scented with the essence of peppermint. But as this seemed to be a better plan, I thought I would try it. I caged the queen to be introduced, and followed his directions to the letter, but what do you think I had? A laying queen in twelve hours? Nay, verily, but a dead queen, and half the bees dead and driven from the hive. Now, Mr. Editor, I think a great deal of my bees, and when, in opening a hive, I carelessly kill one, I am always sorry; but then to see them slaughtered by wholesale, was very cruel to say the least. All the next day, whenever I passed that way, the well bees were driving off those that were crippled or had lost the use of their legs or wings. Besides this, while they were in no condition to repel an attack, the robber bees came in for a share, and I came very near losing them. They were not so drunk but that most of them could crawl round, and only a few of them fell to the bottom of the hive.—Dr. Barnard said, “if they all fell to the bottom it would do no harm.” Now what was the cause of this failure? I could not have smoked them too much, according to his instructions, for nearly all of them could crawl round, when I first opened the hive to let the smoke out; yet it destroyed fully half of them. I do not write this by way of fault-finding, but so that nobody as green as I was, should undertake the same process, and have a like failure.

G. M. DOOLITTLE.

Borodino, N. Y., Sept., 1870.



[For the American Bee Journal.]

The Looking-Glass Once More.

MR. EDITOR:—I cannot think, as Mr. Nesbit does on pages 80, 81 of the last number of the Journal, that either one of his suppositions in regard to the old woman's bees, would do to rely upon. It is not at all likely that a queen so defective as to be unable to fly a distance of two hundred and fifty yards, would ever have been found where this one was.—And as to there being two or more young queens with the swarm, that may be true; but that they went with that swarm in sufficient numbers to divide them on the apple tree, is positively an erroneous idea. The swarm was followed from the apple tree on which a portion of them was first discovered, to the one on which they clustered last, and they did not seek a place so hidden from view as to make it difficult even for me to see that they selected a bare limb on which to settle. They were hived without difficulty, but proved to be bent on pitching their tent in some other section, by leaving the old box hive unobserved the next day.

As to the "knot" theory, I have nothing more to say—than that, if tried right, it will prove equally true with the *inverted glass theory*. But as to the looking-glass having nothing to do with stopping a decamping swarm of bees, it is a grand mistake. In conclusion, I append a portion of two letters which are before me, showing that I am not the only man that places some confidence in a good thing.

"BELLEFONTAINE, *Ohio*, June 25.

"At the time of swarming, I never allow noise of any kind, and have never had a swarm that did not settle. If the apiarian sees his bees rise high and act as though they were going to leave, the reflection of a mirror thrown in among them, is the most efficient means that I know of to make them alight."

"WINCHESTER, *Ohio*, June 21.

“If the apiarian finds that they will not settle, all that is necessary is for him to take a looking-glass and place it in such a position that it will reflect the rays of the sun among the bees, and they will generally settle immediately.”

I write for the American Bee Journal for a purpose different from the object of a teacher, and when I appear as such, will be willing to wear a garb that will not fit *Ignoramus*. But, at the same time, if anything from me serves the purpose of teaching, it will be all right with your brother in bee-culture best known as

IGNORAMUS.

Sawyersville, N. C., Oct. 1, 1870.



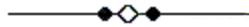
[For the American Bee Journal.]

More About the Looking-Glass.

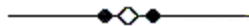
I see on pages 34 and 35, Vol. VI of the A. B. Journal, that Mr. H. Nesbit seems to doubt the efficacy of the looking-glass for stopping a swarm of bees. I would like to tell him an instance, and see if he doubts longer. A near neighbor of mine was at work for me one day, when his wife called him, for the bees were swarming. We went to his house and the bees were just clustering on a tree near by. He got a hive and was going to hive them, when they started to go off. He took a large looking-glass and ran to get up with them, for by this time they had got fifteen or twenty rods from where they had clustered. He reflected the rays of the sun upon them, and they soon began to think of lighting. As there were no trees near by, they began to cluster on his hat; and he, being somewhat afraid of bees, made good time for the house, I assure you. They then settled on a post in the fence near by, and were hived. In about an hour they concluded to try for the woods again; but the looking-glass brought them down once more, and they were hived a second time. In two hours after they started the third time. It being cloudy at the time, they made their escape, as the looking-glass would not work without the sun. Now, was the queen tired or defective, or was it the looking-glass that proved efficient? There were several persons, nearly a mile distant, who saw the reflected rays of the sun, their attention being called from their work by the brightness of the reflection. I am inclined to think it was the looking-glass, instead of the queen being tired or defective. I have since tried it, and never failed to stop a swarm when the sun shone.

G. M. DOOLITTLE.

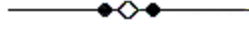
Borodino, N. Y., Sept. 13, 1870.



Pösel says that if a colony has suffered from hunger for twenty-four hours, the fertility of the queen will be greatly impaired, and never be recovered.



All futures are possible to Young Samson. The lion in his path he throttles,
turning his carcass into a bee-hive.



[For the American Bee Journal.]

The Hive Question.

This question has again been revived for discussion in the Journal, and several of our patentees and vendors have made pretty free use of its columns for “blowing” their particular inventions and wares. Prominent among them is Mr. J. H. Thomas; and as I have had some experience with his hive, I wish to have my say about it in particular, and other hives in general. Mr. T. has gotten up a neat and substantial hive, and has admirably adapted the use of frames to the old form of the common box-hive—tall in proportion to its length and breadth. The frames are fixed in their relation to each other, but are as easily moved laterally, when desired, as the frames of any other hive. As there are only eight frames, they can be taken out and examined, when looking for queens, &c., quicker than can be done with hives containing a greater number of frames, and this seems to be considered by some as of great importance. But I do not consider facilities for looking up queens, the most important requisite of a good hive; and I find in the fact of its having so few frames a very serious objection. In order to have the proper number of square inches of comb in a few frames, they have to be made comparatively large, which is the case with these. The frames are so large that, in very hot weather, when the hive is exposed to the sun, and the combs are full of honey, they break down and fall out of the frames, making a very undesirable muss in the hive. I have had this to happen repeatedly, even in his “double wall self protecting hive,” so called, with all the ventilation that could be given it. By the way, he has lately made a change in the ventilation, by enlarging the entrance (an improvement) and by closing the inch and hole covered with wire cloth, in the bottom board, and making another in the back and about an inch above the bottom board. I do not know which is according to “scientific principles,” and whether an improvement or not. It is true this breaking down of combs might be prevented by shading the hive; but the “best hive in America” ought not to require this, as we do not always want our hives shaded. There are several other minor objections to Mr. T.’s hives, but a still more important one will be mentioned presently.

Five years ago Mr. T.'s hive might have been considered a very good one, but "the world moves," and no single department has made greater strides of progress in the last ten years than apiculture. His, and all similar hives, lack one important feature to make it adapted to the present wants of all progressive bee-keepers. No hive should now claim perfection without being easily provided with extra frames for surplus honey to be used in the honey extractor, and these frames should be of the same size as those in the body of the hive. It should be well adapted to the use of the division board, with room at side or ends for surplus frames, or be easily and conveniently converted into a two-story hive, with frames in the upper story the same size as below.—Tall hives with large frames are not well adapted to this purpose. The two-story Langstroth works well. Mr. Gallup's and Mr. Truesdell's style of hives can be easily arranged with additional frames at each end, or on top, or both. Now, I do not say that any and every hive thus arranged is perfect, but that no hive should lay claim to being the most perfect hive made, without being adapted to such an arrangement; for it is important to give for the breeding capacity of the queen, and to furnish a sufficient amount of empty combs for the accumulated workers, and thereby obtain the greatest yield of honey with the extractor, or without it.

Besides "puffs" of particular hives, we have numerous articles on general principles to be observed in their construction—some approving and some condemning the shallow form of the Langstroth hive. In the August number, Mr. J. W. Seay pitches into the shallow hives on general principles and preconceived theories. Now, theories do well enough for fine talk, and are good when substantiated by facts. But facts are the things for the practical man, and one fact is worth a dozen theories. Mr. S.'s theory and deductions therefrom, in regard to the production of early brood, I do not find confirmed in my experience and observation; and the facts of the case warrant a very different conclusion. A tall hive is thought best for wintering out doors, for we know the bees will place their stores above them when there is room. We know, also, that they do not cluster on the honey, but below it, and the heat from them ascends and makes their stores more accessible in cold weather. But how is it with the breeding early in the season? Mr. S. says, "the bees in order to hatch brood as the weather becomes warm in the spring, will cluster at the larvæ end of said combs, &c. Now what he means by the "larvæ" end of the comb, I do not exactly

know. If he intends to say that they cluster at the bottom of the brood comb, so that the heat will ascend and warm up the upper part of the brood comb for the extension of brood, facts do not warrant the assertion; for it is well known that bees do not commence breeding at the lower end of the comb, except in a very rare case, when they have had the hive full of honey and have consumed none or only very little during the winter. As a general thing, they commence breeding near the centre, and frequently in the upper part of the hive. I have known them, in the Thomas' hive, to commence breeding within two inches of the top bar, with plenty of honey at the sides. Now, when breeding is commenced near the top, the extension of brood in a tall hive must be chiefly downward—away from the heat generated in the cluster, instead of towards it. And for this reason, as the warmth of the cluster will be diffused laterally more readily than it will extend downwards, more rapid breeding will be induced in the shallow hive than in the deep one. This accords exactly with the facts of the case. If Mr. S. only means that the bees cluster on the larvæ and around it, he is correct; but this does not alter the conclusion. In stating that the bees will cluster and commence breeding in one end of the low hives, leaving the other end empty and cold, Mr. S. does not fairly state the case. They generally cluster near the centre of the hive, and the heat will radiate towards both ends.

But, we have had enough of theory. How stand the facts? I have had Mr. Thomas' hive—one of the best of the tall ones, and the Langstroth hive, side by side, for several years. Last winter I prepared eight of each kind for wintering on their summer stands, somewhat similar to the plan recommended by Mr. Langstroth. In the latter part of the winter one colony in a Langstroth hive was lost, not from any fault of the hive, but from my carelessness. At the opening of the spring, a thorough examination was made of each hive, with the following comparison: *First*—loss of honey was about alike in each kind; some of each had nearly exhausted their stores, while others of each kind had more than enough, so that when equalized all had plenty. *Second*—loss of bees: In the Langstroth hives this was light. In four of them a spoonful of dead bees could not be found. The other three had a few dead bees. In one of the Thomas' hives no dead bees were found. In two others not a great many, but more than in the worst of the Langstroth hives. The other five had a great many dead bees. The colonies were much reduced—one to a mere handful, with frames and hive

badly soiled with their discharges, had to unite it with another hive. The T. hive that had no dead bees, was in a fence corner, nearly buried in snow all winter. *Third*—mould on combs. In all the Thomas' hives there was more or less mould, except one. No mould in any of the Langstroth hives. *Fourth*—quantity of brood. *Decidedly the most in the Langstroth hives, at the time of the examination, and it increased faster, and they swarmed earlier than the tall hives.* My first swarms came from the flat hives every season. It may be said that the colonies in the flat hives, having lost only few bees in the winter, were stronger and would generate heat and naturally increase faster, and swarm earlier from this cause. I grant it; but one of the tall hives lost no bees, and was very strong, and yet did not breed as rapidly as the other.—I make this statement without favor or partiality. I expected a different result. I have no hives—patented or unpatented, no territory, or interest in any patent, to sell.

I have made a hive on the plan of Mr. Gallup and Mr. Truesdell; which I believe possesses many advantages, and is capable of being used more ways, with the same size frame for all the different styles, than any hive I have seen described. The brood apartment is the plain box of Mr. Gallup—eleven inches wide, fourteen inches deep, eighteen inches long, or as much longer as may be desired. The frames are hung across the narrow way. I have given greater depth and less width than my model, because I wanted to winter out-doors, and because I wanted to use the same frames—in a non-swarmer, with two tiers of boxes at sides. We can use this hive—1st. as a simple frame hive, with large room on top for surplus boxes.—2d. By extending the length to any desired number of frames, frames for surplus honey may be put in each end, for emptying with the extractor.—3d. It can be easily made a two-story hive, with frames in the upper story the same size as in the lower one.—4th. By having movable side-boards, it may be made a non-swarmer on Mr. Quinby's and Mr. Alley's principle, and piles of honey boxes may be put on the sides and top. I have one made this way with thirteen frames, sixteen five pound boxes form the sides, and three twelve pound boxes on top, all enclosed in a suitable case. This is made somewhat like Mr. Alley's hive; but I think is better than his. To avoid one extreme—the flat form, he has gone to the other, and has his hive too tall and too narrow. From all that I have read from our best German and American writers on the subject, I think I have hit the “golden mean” of

width and depth. The great beauty of it is that the same frame can be used in all the different styles; and that we may have a variety of hives with but one size of frame.

I call this hive, with its non-swarming and box arrangement, the “QUINQUEPLEXAL-DUPLEX-COMBINATION-NON-PATENTED-SUPERFLUOUS-HONEY-PRODUCING-HIVE.” It is said “there is nothing in a name,” but if I could only get friend Price’s “*Reversible-Revolvable*” attachment, with the privilege of adding the name, there would be considerable improvement in adopting this compellation for the modified arrangement.

THADDEUS SMITH.

Pelee Island, Ontario, Sept. 10, 1870.



[For the American Bee Journal.]

The Thomas Hive.

Mr. EDITOR:—I wish, with your permission, to correct some few errors which have appeared in the Journal with regard to the Thomas hive in Canada.

Mr. J. H. Thomas, in the July number of the Journal, says—“It is the principal hive in use in Canada.” Again, in the correspondence of the Bee Journal, September No., page 71, Mr. H. Lipset says—“The Thomas hive is all the go in Ontario.” How is it that men will make such extravagant statements? Now for a few facts, as the bee-men say.

One of my neighbors, an intelligent and scientific bee-keeper, having been bred to the business, received a hive from Mr. Thomas, and after giving it four or five years’ trial, says he would not use the hives if he could get them for nothing.

A Mr. Conger, of this county, whose son was an agent for the Thomas hive, told me lately that he had thrown the Thomas hive aside, in favor of a hive similar to Langstroth’s shallow form.

Mr. Walter Taylor, of Fitzroy Harbor, Ontario, formerly an agent for the Thomas hive, wrote me last winter that he would get his bees out of the Thomas hive as soon as possible, as he had found the shallow Langstroth hive was “just the thing.”

I know of no person, making bee-keeping a “business,” who uses the Thomas hive. After all, the Canadian bee-keepers ought to feel proud of having a man among them who has produced the “best bee hive in America.” Where are Dr. Conklin, D. L. Adair, and J. M. Price with his revolvable, reversible—and so on to the end of the chapter? Echo answers—nowhere!

This has been a good year for bees in this part of Ontario. Yet a man living five miles from here, and using the Thomas hive, says it has been a very

bad season.

I commenced in the spring with forty-five hives, several of them being very weak from want of honey. I now have eighty-seven good stocks and sixteen hundred (1600) pounds of box honey, besides about ten frames full. Two stocks that did not swarm produced eighty-five (85) pounds each, of box honey. My first swarm of the season, which came off June 13th and was put in an empty hive, stored sixty-six (66) pounds of honey in boxes, besides losing a frame of honey which melted down with the extreme heat which prevailed this summer.

The foregoing, of course, does not come up to the big stories we read in the Journal; but it is very good for this section of Ontario, and pays very well.

My hives contain nine frames, $16\frac{3}{4}$ inches long and $8\frac{1}{2}$ inches deep, inside. The frames run from front to rear. The hive is similar in shape to Langstroth's shallow form. I obtain earlier swarms and more surplus honey than any other person in these parts using a deeper form of hive. While I put boxes on the top I would not use any other form of hive. I think that Alley's new style of Langstroth hive is the best for obtaining surplus honey in boxes that was ever invented. I constructed two hives last year, as an experiment, similar to Mr. Alley's. One of these gave me the sixty-six pounds before mentioned.

W. Baker, in the September correspondence of the Journal, says that his bees swarmed without making any preparation. Many of mine did the same thing this summer. In opposition to this, on examining a hive five days after a swarm left it, I found a laying queen, and from the number of eggs I saw, I should think she had been laying twenty-four hours at least.

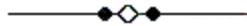
In looking over the Bee Journal, I am surprised to see that so many bee-keepers still use a pan of chips, old rags, rotten wood, &c., with which to smoke their bees. I use a pipe, which for convenience and efficiency, I think cannot be surpassed, notwithstanding Mr. Thomas to the contrary. It consists of a tin tube, six inches long and one inch in diameter, having a funnel soldered to the inside, about $1\frac{1}{2}$ inches from one end, as shown in the annexed figure:



The funnel or cone is punched full of small holes. Into each end of the tube a bored plug, *a* and *b*, is nicely fitted. The plug *b* is cut so as to be easily held between the teeth. To get the smoke, draw out the plug *b*, fill the space *c* with some combustible material, then with the plug *a* in the mouth, it may be lighted with a match, like a common pipe. When lighted, insert the plug *b* in its place, and blow away. I have used cut tobacco till lately, but now find dry corn silk much better. The advantage of this pipe is, that it can be held in the mouth, and the smoke directed where it is wanted, while the hands are free to operate with. This is a great convenience, especially in taking off boxes.

GEORGE CORK.

Bloomfield, Ontario.



[For the American Bee Journal.]

Shallow Hives, or Deep?

Mr. EDITOR:—In the September number of the Journal, Dr. B. Puckett criticises an article of mine in the July number, and asks me to explain wherein the shallow Langstroth hive is lacking.

When I wrote the article referred to, my object was to show that the shallow hive could be altered to a different form, and that those who were using it, and considered it too shallow, need not throw their hives away. I said it was *not* a good hive for wintering in the open air, or for early spring. I did not think it necessary to give my reasons in detail, why it was not good; for that matter I considered had been already fully discussed in the Journal. But as Dr. P. requests it, I will explain.

For wintering in a cellar, the hive is perhaps good enough. But I do not want to be *obliged* to house my bees. Sometimes I have plenty of room in the cellar, and sometimes not. If the hives are of suitable form for wintering in the open air, I can let them remain out, when it is not convenient to carry them in. But the great objection to them is in early spring. Dr. P. asks if it is the fault of the hive that the *old* bees die off, or that bees are destroyed by cold winds? Of course it is not. But if a swarm is not breeding enough to make up that loss, there must be a fault somewhere. When we take bees from the cellar, we expect that they will have brood in all stages, from the egg just laid to young bees just gnawing out. We expect too that the queen will continue to deposit eggs, even more rapidly, because of the excitement produced by the bees flying, and especially if they are fed rye meal, as mine always are. I said, after they had been out *a month*, there appeared to be fewer bees than when first carried out. We expect a loss the first day or two after taking them out, but soon afterward, the bees should be increasing; and at the end of a month, which brings it into April, there should be a decided increase. In deeper hives, according to my experience, it is so; and the deeper the hive the greater the increase.

The reason why the shallow hive is not good for early spring, as I understand, is this: as soon as severe weather is past, we want to confine the animal heat as much as possible to the hive, that the bees may breed rapidly. Consequently we shut off all upward ventilation. The coldest part of a hive is near the entrance and so along the bottom board. The farther the bees get from the bottom, the warmer they find the temperature. These hives being so low, before the bees get out of the way of the cold air coming in at the entrance, they are bumping their heads against the top. And, instead of spreading the brood in a circle, which is the best form to economise heat, they are obliged to carry it along horizontally, and after all work at a disadvantage.

In a tall hive they can draw up and get well out of the way of the cold air from the entrance. The top of the hive being small, *the animal heat, brood,* and bees are all compact, and in the best condition for rapid breeding. The faster they breed, the faster they can breed, as there are more bees to keep up the heat; and as it naturally ascends, the smaller the hive is across the top, the more compact the heat will be kept.

A friend, who for some years has been using a very tall hive, after trying for a long time to persuade me to use some of them, finally gave me one in the spring of 1868, and requested me to put a swarm into it. Says he—"You may let it stand anywhere through the winter; the bees will be sure to do well." I have used it, and found that the bees increase in it nearly twice as fast in April and May, as in the shallow hive. The result is the same in his apiary.

Mr. Alley, who at one time so vigorously advocated the shallow hive, has since become convinced of his error, and invented what he calls the new style Langstroth hive. The shallow frames are set up endwise, which gives it extreme depth. In the September number of the Journal, 1869, page 54, he says—"I examined fifty stocks of bees in shallow hives last spring (and many of these were larger colonies than any I had); but none of them had as much sealed brood as mine."

When he first got up this hive, and before any of them had been used, a friend of his had one, and was requested by Mr. A. to show it to me and get my opinion upon it, not letting me know where it came from. I refused to

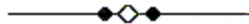
express an opinion, except on the point of wintering, in which I considered it could not be beat.

The great depth of combs, together with the protection given by the outer case, makes it one of the best hives for wintering that I have seen. It has a large amount of box room for surplus honey, which is needed for a swarm that has been well wintered, and that has increased well during the spring. But let him just turn the frames down to a horizontal position, making it a shallow hive, and I will guarantee that one-half of the box room will be ample.

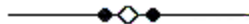
I have attempted to explain wherein the shallow hive is lacking, and now have a favor to ask of Dr. P. He says: "The Langstroth hive could be made deeper very easily, without Mr. R.'s patchwork." Will he tell us how it can be done, and still retain about the same number of cubic inches?

CALVIN ROGERS.

West Newberry, Mass., September 10, 1870.



Honey is the most elaborate of all vegetable productions.



[For the American Bee Journal.]

Wintering Bees.

We republish the following from the A. B. J., Vol. IV., page 109, at the request of a number of new subscribers. We regard it as probably the least troublesome and most successful mode of out-door wintering yet devised.

It is settled beyond a doubt in my own mind, by the experience of others as related in the BEE JOURNAL, and by my own experience for several years in the apiary, that bees to winter well, must have sufficient ventilation to carry off the excessive moisture which accumulates in well stocked hives. This moisture arises partly from the exhalations from the bodies of the bees, but mostly, I think, from the surrounding atmosphere, which constantly holds in suspense a greater or less amount of moisture, according as its temperature is higher or lower. The warm atmosphere of the hive is capable of holding a considerable quantity, until it is condensed by coming in contact with the cold walls of the hive, at some distance from the cluster of bees. There it condenses, first into minute drops of moisture, and afterwards, if the cold increases, into frost. The constant accumulation of the quantity, by repeated thawing and freezing in a hive that has no efficient means of ventilation, gradually encroaches on the space occupied by the bees, finally reaching those on the outside of the cluster. These grow benumbed, cease to eat, lose their vitality, grow cold, the frost forms on their bodies, and they die where they stand. The frost continues to penetrate the cluster, if the cold weather is prolonged, until finally the last bee dies covered with frost. The warm days of spring then melt this frost, and on examination, the whole mass of bees are found dead and as wet as if just dipped from a basin of water. I found one hive in that condition last spring. The entrance to this hive was left open, but the honey-board was left on tight, without any upward ventilation, as an experiment. All my other colonies wintered well on their summer stands, having their entrances open three or four inches wide, and the front and rear openings in the honey-boards (half an inch wide, and extending the whole length of the hive) uncovered, but the middle opening closed.

For the coming winter I have adopted Mr. Langstroth's plan with some modifications. I shall omit the outside covering of the hive, believing that it is better to have the hive of a single thickness of board, say seven-eighths of an inch, in order that the heat of the sun may easily penetrate it, and warm up the hive almost daily, thus giving the bees an opportunity to bring to the central part of the hive fresh supplies of food from the outer combs. This plan *may* lead to a somewhat greater consumption of honey; but if a swarm of bees will give its owner from fifty to one hundred pounds of surplus honey in a season, as mine have done the past summer, he ought to be entirely willing to have them eat all they need during the winter. At all events, one of two things must be done, to winter bees successfully, in addition to their having a supply of food and thorough ventilation—they must either be kept in a repository where frost cannot enter, as a cellar, trench, ice-house, or the like; or they must be put where the sun can warm them up occasionally.

I have removed all the honey-boards, placed two one-half or three-quarter inch strips across the frames, and covered the whole top of the frames with any old woollen garments that could be found about the house.³ These need no cutting or fitting. Pack them in as you would pack a trunk, (the roof or cover of my top box is movable, and I like it much better than the old plan of having it nailed on,) two, three, or half a dozen thicknesses will make no difference. The moisture will pass through as readily as the insensible perspiration of our bodies will pass through our bed covering. The hives will remain dry and the bees warm. I have no fear of losing a single swarm the coming winter, although several new ones which I bought are quite weak, owing to the sudden close of the honey harvest a month earlier than last year, in consequence of the drought.

R. BICKFORD.

Seneca Falls, N. Y., Oct., 1868.



[For the American Bee Journal.]

Upward Ventilation.

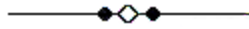
MR. EDITOR:—I once found a bee-tree, with an excellent swarm in it. I cut it down Gallup-fashion, and moved it home, in the month of February. The entrance was a hole, about three inches in diameter, just at the top of the cavity. The tree was a green butternut. I sawed it off, short enough to handle easy, and set it up in the yard. The combs were bright and clean, and there were not over a dozen dead bees in it when found. It swarmed twice in June following, and next winter I stopped up the entrance at the top, and made another within six inches of the bottom, by boring a two-inch hole through the side. All this time I kept the top closed tight. The following winter I came near losing them with dampness and dysentery. Next winter, I closed up the auger hole, and opened the top entrance again. They wintered as nice as a pin—no dampness or dysentery. In April I thought I could still better their condition, by making the entrance smaller, and reduced the entrance to one inch in diameter. Within six days after, I came near losing them with dampness and mould. Experimenting still further, I noticed that the fanners or ventilating bees would, in hot weather, be arranged in this manner—one set at the lower edge of the entrance, with their heads outward; the other set at the top of the entrance, facing inward, driving out the hot air. I then reduced the size of the entrance still more, and found that in a very short time nearly the entire swarm would issue and cluster on the outside of the log or gum. Enlarging the hole to three inches again, the bees would soon return inside and resume work. I kept that log hive four years, and then sold it to a neighbor. Whenever I wintered it with the natural entrance open, there was no dysentery and no unnatural distention of the abdomen; and on their first flight in the spring, they would not even speck the snow.

In wintering bees in the Wellhuysen hive, made of willows and plastered with cow manure, they would never have the dysentery—not the least sign of it. The combs were always bright and clean, and the bees always in as good condition as they were in midsummer. I have wintered bees in Canada, in the old-fashioned straw hive, with the entrance, summer and winter, a two-inch hole in the centre at top; and they always wintered well,

without the least sign of dysentery, even when they would not leave the hive from the 10th of October to the 1st of May—nearly eight months. In that climate they are nearly always confined from the 1st of November to the 10th or 20th of April, or about five months. When I lived there, there was scarcely ever any honey stored after the 15th of August, yet bee-keeping pays in that climate. To encourage our northern bee-keepers, I will say that, according to my experience, there and in the West, I think the flowers secrete more honey, in the same length of time, there than here. Our atmosphere is rather dry, while theirs is moist and humid—just right for the secretion of honey.

ELISHA GALLUP.

Orchard, Iowa.



[For the American Bee Journal.]

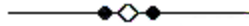
Alley's Improved Langstroth Hive.

MR. EDITOR:—For twenty years I have had experience in bee-keeping, and had within that time as many different styles of bee-hives in my apiary; but, taking everything into consideration, the advantages derived from Mr. Alley's, proves it to be the best I have yet seen. It has the best shape, the greatest amount of animal heat for wintering bees, and as for storing honey, it allows as much room for surplus honey as the largest stock would need.

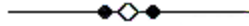
These are only two among the many advantages it presents. Many more might be mentioned. I simply state these, as I consider them the most important. Brother bee-keepers, who are about to purchase, should not fail to give it a trial.

LEVI FISH.

Danvers, Mass., Sept. 10, 1870.



Intelligent practice is very different from blind practice; or, in other words, practice preceded by a sound theory is evidently far superior to practice without theory.—TALBOT.



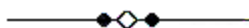
[For the American Bee Journal.]

Ventilating the Gallup Hive in a Damp Cellar.

The cellar of my house is nearly underground. Its size is $38 \times 28 \times 7$ feet, inside measure. The temperature during the winter is usually 38° F., with occasional extremes of 35° and 41° . It is damp, and not specially ventilated. A stairway from the porch and one from the kitchen, furnish all the air; the latter being very much used during winter time. In this cellar I have usually wintered some of my bees, for many years—trying various methods and different kinds of hives, with the result always, till last winter, of more or less mouldy combs. I then had among the lot four strong stocks and Gallup hives. These I had setting up three feet from the ground, with caps and honey-boards removed, and the loose top cover laid directly on the hives; and by means of hard wood wedges pushed in between the lower edges of the hives and the bottom-boards, and also between the upper edges and the top covers, I gave them one-eighth of an inch air all round the hives, above and below, except six inches in length at the entrance, where I gave them one-fourth of an inch, so that the bees could get out. In this condition the hives were left all winter. The bees remained very quiet, humming almost inaudibly, and paying little attention to the light of a candle which was carried in many times a day. Scarcely any came out to die; and not over half a teacupful died in each hive. They consumed comparatively little honey, and when the hives were examined after being set out in the spring, the combs were all dry and free from mould. In my experience absorbents used on a hive in a cellar have always caused combs to mould. Who would think of laying on top of his hives a damp straw mat, or a pile of damp corncobs? And yet it is all about the same thing. Give the proper amount of air, and let it pass off unobstructed. I shall try a larger number of hives the coming winter. Many thanks to Gallup.

HENRY CRIST.

Lake P. O., Stark county, Ohio, Oct. 4, 1870.



[For the American Bee Journal.]

Bee Hives, and Shipping Honey in Frames.

There has been much said on hives in the columns of the Bee Journal. Some are said to be too deep, and others too shallow. But after all, profit in dollars and cents is the great object; and to secure this in the shape of surplus honey, three things are requisite—*first*, strong colonies of bees; *second*, a good season with plenty of pasturage; and, *third*, the placing your surplus honey boxes or frames as near as possible to the brood in the main body of the hive. There are two ways to accomplish this: *first*, by using the shallow form of hive, with frames say seven or eight inches in depth; and, *second*, by using the side gathering or storing hive. I prefer the latter, with frames twelve inches deep; and this for three reasons. *First*, if the apiarian has no repository for winter quarters, his bees are right in these for wintering in the open air. *Second*, the brood and cards of honey can be so adjusted as to bring the former next to your honey boxes, if necessary; as we never want more than one full frame of honey between the brood and the surplus honey boxes or frames. *Third*, in the manipulation of colonies there is no comparison between the side storing hive, and the top storing. With the former, when the lid is removed, we have access to the frames, without the intervention of surplus honey or other boxes. Top-storing hives are now behind the age.

Those using shallow frames must, in this latitude and climate, have a house for wintering their colonies, and when bees are removed to their summer stands in the spring, the lid that covers the second-story or surplus honey chamber, should fit on the brood chamber, that the honey chamber may be left off till the time comes for placing surplus honey boxes on your hives. By this means all the heat rising from the bees is secured and diffused through the main hive or brooding chamber for hatching the eggs; and the bees multiply as rapidly for aught I can see, and swarm as early as in the twelve inch frames. I have used one hundred shallow hives, with frames eight inches in depth, for three years; and when I suffer them to throw off natural swarms, they swarm as early, sending off as many and as large swarms as taller hives.

In 1869, I had gathered six thousand pounds of fine surplus honey in frames in the top receptacles of my shallow hives. A large proportion of this I shipped, in the frames, to C. O. Perrine & Co., Chicago, Ills. They paid me twenty-five cents per pound for it, frames and all. Should any honey raisers in the West wish to sell to a good man, I should recommend them to Mr. Perrine. I have trusted him with quite large amounts at a time, and always found all right at settlement day.

SHIPPING HONEY IN FRAMES.

To do this properly and safely make the box or case in which you ship only wide enough to receive the length of the top bar of your frames, and one and a half inch deeper than the depth of the frame. Make the case tight and pitch the inside with rosin and bees wax, so that the leakage of the combs will not be lost.

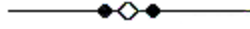
In packing the frame honey, first pierce the projection of the frames through with an awl, invert it and place in the holes one inch finishing nails, then place the top of the frame down and crossways in the case, and with a tack hammer drive your nails. Place the next frame by the side of this first, corresponding as built in the hive, if it can be; and place them so as slightly to touch. In filling the last end of the case, place an iron rod on the head of the nail to drive it, as you cannot play the hammer.

When the case is full, take two strips (common lath) just long enough and wide enough to fill the case tightly from end to end, and cover the ends of the frames and fit tightly against the sides of the case; drive an inch nail through the strips in the end piece of each frame, and the frames will be perfectly solid.

I shipped from one to two hundred pounds in a case, in this manner, and Mr. Perrine tells me the average was not over two frames broken down per case, and no loss from leakage, the boxes being pitched inside.

A. SALISBURY.

Camargo, Ills., Sept. 6, 1870.



[For the American Bee Journal.]

The New Smoker.

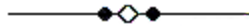
I introduce to the notice of bee-keepers a new smoker for bees, believing it will be pronounced the best, until a better one is found.

It will be found the best for ease of lighting, and to retain fire, and as burning with equal facility, rotten wood, old rags, or a combination of wood and rags; and it will not annoy the operator every few minutes by going out.

To make one, procure a piece of wove wire; I use very fine wire cloth, but suppose that a coarser article will answer. The piece should be twelve inches wide and from twelve to eighteen inches long. Take of old rags a sufficient quantity to make a roll about 2 or 2½ inches thick and twelve inches long. Roll the rags evenly and firmly together, and then lay them at one end of the sheet of wove wire, and roll the wove wire over them pretty tightly, and bind with wire. Light at one end with a match; and your smoker, if nicely made, will burn from two to four hours. Or if it be only half filled with rags, then fill out lightly with damp rotten wood, and you will have a big smudge.

JOHN M. PRICE.

Buffalo Grove, Iowa.



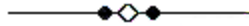
[For the American Bee Journal.]

Reply to Mr. Worthington's Inquiry.

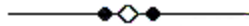
MR. EDITOR:—I see in the June number, page 264, Mr. Worthington asks how to examine bee stores, &c., in the American hive. Here is the way I do. Remove the cap and honey box; blow a little smoke through the slot in the top bar of frames, to quiet the bees; remove the movable side, and with your pocket knife, you can easily run the blade between the top bars, loosening them; lift out the frames, placing them in a skeleton frame made to hold them; and in this way you see *exactly* the condition of your bees. In returning the frames to the hive, you have only one place to watch to prevent killing bees, that is the top.

J. W. SALLEE.

Pierce, Mo.



If asked how much such contrivances against the moth will help the careless bee-man, I answer not one iota; nay, they will positively furnish him greater facilities for destroying his bees. Worms will spin and hatch, and moths will lay their eggs, under the blocks, and he will never remove them. Thus, instead of traps, he will have most beautiful devices for giving effectual aid and comfort to his enemies.—*Langstroth's "Hive and Honey Bee."*



[For the American Bee Journal.]

Bees in Bennington, Vermont.

MR. EDITOR:—The season in Bennington has been very good for bees, that is, considering that they were in poor condition last spring. Many colonies died last winter in this town, and I should think it safe to say that one half our bees then perished for want of honey. I was not at home in February to attend to mine, and lost five colonies before I was aware of their being so short of supplies, which I discovered only after losing my best stock of Italians. It was quite warm in January, and one day was so like spring that I carried my hives all out, and for a couple of hours it seemed like swarming time. The weather was so mild that my bees began to breed considerable, and so used up their honey. When I removed the dead bees from one of my hives, I found brood in three combs sealed over, a spot as large as my hand in each, besides eggs and larvæ.

February was very cold, and a terror to light swarms. I set my hives out again the last of March, and had then only fifteen stocks. Three of these I united with others, thus reducing the number to twelve. One of these got discouraged, and tried to form a partnership with another colony, but got killed in the operation. Thus, by the first of May, I had only eleven colonies remaining, and they were very weak. I fed them every day till I began to see they were getting stronger. Then, thanks to the Bee Journal, I knew enough to double their feed as they increased in numbers and the hives in weight of brood, for they could not of course get much honey till the first trees blossomed. The weather then became warm and pleasant, and the bees got a good start in life, so that when clover and red raspberries bloomed, they were soon ready to march out and take a limb of a tree on their own account. I soon had twenty-five swarms and began to think hives and all would swarm. Besides those we hived, four swarms took the wings of the morning. By the way, a great number of swarms ran away this year to the woods. I found a small swarm about three miles away from home. They came over a barn I was painting, and clustered near by. I hived them in a powder keg, and carried them home at night.

I have taken two hundred and twenty-five (225) pounds of box honey from my bees, besides ten six pound boxes partly filled, of which I take no account. I have twenty-one hives to winter. They are very heavy, too heavy, I fear, to winter well; but hope for the best. Bees within half a mile of mine have not done anything at all; because they had no care or feeding in the spring, and when summer came they were merely ready to begin their spring's work. I think it pays to feed bees as well as other stock.

I have only two swarms of black bees, and some hybrids, the rest are pure Italians. I received two queens from Mr. Cary this season, and inserted them all right. They were, to all appearance, accepted and owned for four or five weeks, when one day I found one of them thrown out dead on the bottom board; and if it had not been for the Bee Journal on the superseding of queens, I should not have known what the trouble was. The other is all right so far, and the young bees from both queens are beauties. I never saw finer, and am well satisfied with them. My bees are all descendants of Mr. Cary's stock, and another year I shall get some more from him and other breeders, to avoid breeding in and in.

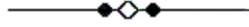
I have never yet seen a honey extractor at work, but there is one within a few miles of me and I am going to see it. If it proves to be the one thing needful in my case, I shall go for one another year.

I have procured some of the Rocky Mountain bee plant seed from Mr. Green, and if it is good, as I have no reason to doubt it will be, I shall let you know all about it.

The season has been quite favorable here, not as dry as it was in some places; and our crops are very good, with an abundance of fruit. Taking every thing into consideration, I am well satisfied with my bees and their labors last summer. When I bought my bees, a man in the same business blowed a good deal and said it would not be a great while before I would run out with my Italian bees and wintering in the house. Last year (1869) he had in the summer sixty-six colonies. He fed two barrels of sugar this spring, as he says, and now has twenty or twenty-one colonies. Who has run out? I fed half a barrel or one hundred and twenty-five pounds of sugar. He don't "fool away his money for Bee Journals, nor Italian queens."

C. H. BASSETT.

North Bennington, Vt., Oct. 5, 1870.



[For the American Bee Journal.]

The Season in Massachusetts.

After reading the various accounts in the Journal as to how bees have done in other parts of the country, I think it will not be out of place to let its readers know what has been going on in Massachusetts, or rather in a part of that State.

About May 20th our bees commenced to collect honey rapidly, and from that time to June 7th, honey was very abundant, and I never saw bees put into the hives and surplus boxes faster. From June 7th until July 1st they did very little. In fact we had then ten days in succession when no honey was collected; and by the 1st of July pasturage failed altogether, as it generally does here in New England. I never knew bees to put honey into boxes later than July 12th, and that for only one year, since I have kept bees.

Perhaps it will be new to some of the readers of the Journal to know the fact that bees do not collect honey here, in Essex county, as a general thing, later than the first week in July; and this season they did not work later than the last day of June. Very little honey was put into boxes between June 7th and July 1st. Had the season held out as it gave promise in May, honey would have been plenty in Massachusetts.

I have a few hives that did very well, considering how short the honey harvest was, and to let some of your readers know that Alley can raise honey as well as queen bees, I enclose a short report that was intended to be shown to the "Honey Committee," at the Essex County Fair; but as I was the only person who exhibited bees or honey (except four small boxes by Mr. Gould, of Ipswich,) I did not submit it. Of course Alley got the highest "premium," under such circumstances. I suppose if I say that the stock that did best was in one of Alley's hives, some one will think that this article is meant only for an advertisement. Well, I cannot help that; so here goes for the report, and all who do not want to believe it, can accommodate themselves in that line, and I will find no fault. I do not, myself, believe more than only just what I think is true, even when I see it in the A. B. J.:

HIVE No. 1, filled sixty-eight 2½lb. boxes, and cast one small swarm. The honey was sold at thirty-five cents per pound, box and all. Weight of boxes and honey 170lbs.; weight of the sixty-eight boxes empty 34lbs.; net amount of honey stored 136lbs., which, at 35 cents per pound,

is	\$47 60
One young swarm	<u>3 00</u>
Whole amount	\$50 60

HIVE No. 2. This was a stock transferred from a box hive to a movable comb hive, May 26th, 1870. It filled thirty 3lb. boxes, and the honey was sold at thirty-five cents a pound, without including the boxes. Net amount of honey stored 75lbs.; which, at 35 cents per pound, is \$26 25

HIVE No. 3, filled two 15lb. boxes, and cast two swarms. The first of these swarms filled a new hive, from which I have taken twenty-five pounds of honey, and it now has enough to winter on, without feeding. The second swarm I used to rear queens, and it was worth five dollars to me.

Value of first swarm	\$7 00
Value of second swarm	5 00
55lbs. of honey at 35 cents per pound	<u>19 25</u>
Whole profit from Hive No. 3	\$31 25

The profit from these three hives is one hundred and eight (108) dollars.

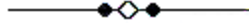
I omitted to say that I took twenty-five pounds of honey from Hive No. 2, as late as August 20th. That hive now has honey enough to winter well.

Since September 20th, the bees have put in a considerable amount of honey, but not in surplus boxes. Even my nucleus hives put in enough from September 20th, to keep them—making a saving to me of twenty-five (25) dollars.

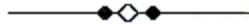
If other bees in this vicinity have done as well as mine, few colonies will starve in this county next winter. My article is getting long. I will stop just here.

H. ALLEY.

Wenham, Mass., Oct. 3, 1870.



Virgil recommends the hollowed trunk of the cork tree as a hive, than which no material would be more admirable, if it could only be easily and cheaply procured.



[For the American Bee Journal.]

Bees at Binghamton, N. Y.

MR. EDITOR:—Having gained so much instruction and pleasure from the perusal of your valuable paper, I think it no more than right to send you a report of the season's operations here. But there are so many of your contributors so much more successful, that my account will appear tame in comparison; yet when compared with what has been done by my box and Kidder hive neighbors, it seems to be quite a success.

The season has been favorable in this locality, though rather dry for many crops, yet honey was more or less abundantly yielded all through the season. The weather has been such that the bees could gather honey almost every day, from the first of May until the present time.

We placed ten (10) swarms in the cellar in the fall of 1869, all of which wintered in good condition and came out strong in the spring. Four of them were Italians, and six blacks, seven in movable frames and three in box hives. Those in the box hives were transferred in April; the black queen killed about the first of June, and young, fertile Italian queens of my own raising substituted for them. One hive was broken up into nuclei in May, and also the first swarm. We have run from six to ten nuclei all through the season, to obtain, if possible, a pure queen for every hive; but we have not succeeded in getting all full marked workers in more than half of the stocks, as our box hive neighbors kept us flooded with common drones.

We have taken this season, as surplus, eleven hundred (1100) pounds of honey—eight hundred (800) pounds being comb or box honey, and three hundred (300) pounds extracted; and have increased our stock to fifteen (15) full swarms. Besides the surplus, we have forty Langstroth frames filled with comb and honey, averaging two pounds each. This is not counted as surplus, but reserved for next season's operations.

After transferring last spring, and cutting out drone combs, our hives lacked from one to two frames each, from a full complement. Having constructed a

slinger this season, we are enabled to lay by a goodly store of combs for future use.

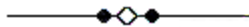
Our best stock gave us twenty-four six pound boxes, weighing one hundred and forty-four (144) pounds, and twenty-five (25) pounds of extracted honey; besides ten frames of brood and honey, taken from the body of the hive at different times in the season and replaced with empty frames. It is now in good condition.

This is the first season that we have practiced non-swarming on the true principle of making box honey, and had we had the knowledge and experience that we now have, we are confident we could have attained still more favorable results. We are no friend to increase, and would never increase more than is absolutely necessary. Nor can we understand how some men are so well satisfied with a large increase and a small amount of surplus. Yet we have not seen any feasible plan put forth whereby a large amount of surplus can be made without a slight increase.

After having tried both kinds to our entire satisfaction, we think we can get as much profit, and far more pleasure, from one Italian swarm in a Langstroth hive, than we can from twenty-five (25) swarms of black bees in box hives.

J. P. MOORE.

Binghamton, N. Y., Oct. 3, 1870.



[For the American Bee Journal.]

Bee Report from Champaign Co., Ills.

MR. EDITOR:—I write to let you know how bees have done here, this season. I had last spring fifty-one (51) stocks, nearly all in my own hive with frames, and on the top four glass boxes, holding ten pounds each, box and all. I sold two stocks for thirty-six dollars, and they earned for the man who bought them one hundred (100) dollars, in swarms and honey.

During the blooming of the trees in the spring, bees had a week to gather honey. Then they did not get any more until the white cover blossomed, and we had a rain on the 10th of June. From that time until the 25th of June bees did splendid; but after that to the 1st of August, they did not collect as much as they consumed. Then we had the fall flowers, and they have done very well.

I bought ten swarms on the 23d of June, but before they commenced work forage failed. I fed them and four of my own late swarms; one hundred pounds of sugar and two gallons of honey. I then stopped until the first of September. Then I fed them over one hundred pounds more of sugar, doubled up three colonies and broke up two. So I now have seventy-two (72) stocks, all of which I think will winter.

My bees have made about 800 or 900 lbs. of honey. To strengthen the weak ones, I took off boxes full of honey and bees, and gave them to weak swarms. Thus they got bees and honey at the same time. In doubling swarms, I open both hives and take five of the lightest frames from one, and five of the best from the other, put them in and brush all the bees out, and they will not fight.

Bees have done better in the country than in the village, as our village is nearly overstocked. The Spanish Needle is a good honey-producing plant; also a tall flower called Wild Artichoke.—It has been very dry here; but rains have gone in streaks. Two or three rains come in the right time, would have been worth a thousand dollars to me. The white clover dried up early.

The bees visited the groceries and were lost by thousands. My bees are nearly all Italians, which I consider the best.

I gave a description of my hive in the Journal, last year. Every one uses it here. It costs about four dollars, and can be made for a little less.

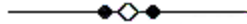
We have had no frost yet, and the bees are collecting honey still, and will do so as long as the Wild Artichoke lasts. I feed my bees by taking off one of the boxes, and put on a saucer with some pieces of comb in it. Then dissolve sugar and fill the comb and saucer. They will take it up every night. Feed till you get them heavy enough.

I divided ten swarms, and they did well, though I divided them too late in the season. If one is going to divide, it should be done early.

Last year was a splendid season for honey. Thirty-two weak stocks gave eighteen swarms, and twenty-six hundred pounds of honey.

DR. H. CHAFFEE.

Tolono, Ills., Oct. 3, 1870.



[For the American Bee Journal.]

White Clover crop.—Buckwheat yielding no Honey.

MR. EDITOR:—I once more take up my pen to advocate bee-keeping. As I said in my last article that my apiary was increasing, I have now ten new swarms from eleven old colonies, and I am every day expecting some second swarms to issue, as queens in the hives that sent out swarms, can be distinctly heard uttering the word “peep! peep!” and according to more able apiarians than myself, that is the true sign that second swarms will issue in a few days, if the weather be favorable.

The other morning I was out where my bees are. I suppose you have a strong idea of what I saw, when I raised up one of my stands. There were a half dozen of the fattest full grown moth worms almost any one ever saw. They were lying back in all their glory, after gorging themselves with the rich feast on which they no doubt had luxuriated. I made short work of them, however. Those round, plump, greasy-looking fellows seem to think, from all appearance, that they are lords of creation. But I soon dislodged them from their snug quarters, by means of a sharp-pointed iron bar made for the purpose. “They slept rather late that morning, and were caught up with.”

The piping of the young queen was something new to me. I told some of my bee-raising friends of it, and they hooted at me, calling me a deceiver and impostor. I referred them to Mr. Langstroth’s book, and Mr. Quinby’s, and told them that they should subscribe for the American Bee Journal, or even read it, and they would find that what I said in regard to the young queen’s piping, was strictly correct. My friend Mr. K. (whom I converted) in a conversation with Mr. S., asked him why he did not take the American Bee Journal. “Why,” replied he, “they can print anything in a paper, and there are fools enough to believe it.” I have known Mr. S. for about fourteen years, and know that he has had bees all that time. Yet he has not any more stands now, than he had ten years ago. (It is no wonder.)

The honey product of this season seems to be good. Bees are storing great quantities of surplus honey. The weather has been very favorable for honey-gathering, for the past six weeks. White clover has been in bloom for the last fifteen days, and will probably continue till the middle of July. From it the best honey is gathered. In the spring the early flowers were cut off by sleet, which fell about the 18th of April.

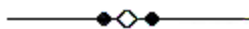
I am now preparing to sow a large field with buckwheat, exclusively for my bees, though some writers in different papers state that the bees do not get any honey from this plant. Whether it is a honey-producing plant or not, the bees seem to visit it as regularly when in bloom, as if there was something about it they are very fond of. Perhaps I can throw some light on this subject. Last fall I had three hives of bees, that came late, while nearly all the other flowers were exhausted, and buckwheat was their only resource for supplies for winter. They worked like white-heads, as long as the blossoms lasted; and after that went through the winter safely, though they were weak the following spring.

I will now give my opinion on ventilation, for the benefit of Mr. A. Green. My mode is as follows: I leave the summer entrance open, and also upward ventilation all winter. I have always, heretofore, wintered my bees in the open air. If Mr. Green uses hives with movable caps, he can close the summer entrance and take off the surplus honey-boxes, substitute straw or fine shavings in their stead, and replace the cap as before. This is the best way that I have yet tried. I intend this for winter. In summer I give them all the ventilation needed—that is, I leave all the ventilators open.

I have drawn out this article longer than I intended, and close with greeting to all bee-keeping friends.

T. H. WOODY.

Pleasant Valley, Mo., June 18, 1870.



[For the American Bee Journal.]

Honey-producing Plants.

MR. EDITOR:—Not having much to do, at present, I thought I would give your readers an account of my observation and trial of the different kinds of honey-plants around us here. It may be of some service to new beginners, as I have tried all kinds I could hear of and procure, that were reputed valuable for producing honey.

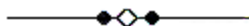
Among the best are Alsike clover, Melilot clover, White Dutch clover, Borage, and Buckwheat. These, with us, just fill out the season from June to October.

The plants named in the following list, I do not consider of any account here, for honey, viz.: White Mustard, Black Mustard, Rape, Chicory, Mignonette, Lucerne Clover, and the Rocky Mountain Plant. Kale did not come into blossom, and I cannot speak of its value as a honey-yielding plant.

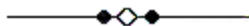
R. MILLER.

Rochelle, Ill.

Some of the plants named as of no value for bees, are highly praised, in other localities.—ED.



I once met with an individual whose breath, shortly after he was stung, had the same odor with the venom of the enraged insect.—*L. L. Langstroth.*



[For the American Bee Journal.]

The Rocky Mountain Bee Plant.

(POLANISIA PURPUREA.)

MR. EDITOR:—About the middle of August, by invitation of Mr. Alfred Green, of Amesbury, a friend and myself visited his place to see the bees work on the Rocky Mountain bee plant. We arrived there about eight o'clock in the morning, and found the plants swarming with bees; one, two, and in some cases three bees upon the same flower.

Mr. Green informed us that they were still at work on it, the day before, at seven o'clock in the evening. It was amusing to see them gather pollen from it while on the wing, the stamens extending so far out that they could not reach them after alighting on the flower.

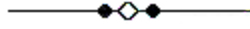
The plant was growing on a rather light soil, not highly manured, and stood from two to three feet high, branching out in all directions. Planted in the spring, it comes into blossom soon after the white clover disappears and continues until killed by the frost. If planted in the fall, as Mr. G. says it can be, it would blossom much earlier. I think this is the best plant to cultivate for bees, as it fills a vacancy, (in this locality) between the white clover and the fall flower.

Alsike clover I have raised, commencing in 1860; and find that, on my soil, bees prefer it to white clover. But as it begins and ends blossoming at the same time with white clover, it is not of so much value for bees, as it would be if it came a month or so later.

As the seed of the Rocky Mountain bee plant is valuable for poultry, and probably for swine and other farm stock, when made into meal, it would perhaps pay to raise it for the seed alone.

CALVIN ROGERS.

West Newbury, Mass., Sept. 12, 1870.



[For the American Bee Journal.]

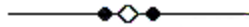
Silk Weed or Milk Weed.

Well, Mr. Editor, I saw in the Bee Journal for July something concerning the injuriousness of the silk weed or milk weed. After reading the article it struck me that there was some of this weed in the vicinity of my apiary, and next day set about to search for it. On going out west, on the low ground on the prairie, I found ten flowering stems of this weed, and seven of the ten had bees fastened on them. Some of these bees were dead, and some still living, though they could not leave the flowers, being fastened in them by their hind legs. The bees seemed to have been gathering honey.

Last Monday, as I was going to a neighbor's, I saw one of these flowers, three quarters of a mile from my home. I stopped to see if I could find any bees on it, and found an Italian just alive. I am glad there are not many of this species of plants in this neighborhood.

R. MILLER.

Rochelle, Ills.



[For the American Bee Journal.]

Honey Dew.

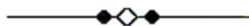
MR. EDITOR:—I have at last caught the chaps that rain down what is called honey-dew. In localities where the common willow grows, I found the most. On the Missouri river bottom, which is literally covered with willows, I find in June and July they are covered with small insects, which at a certain age get wings and fly off in large swarms, going for miles. Sometimes they will stop in the air, over some trees, and fly around in a circle for an hour. If you get them between your eye and the sun, you will see them discharging the so-called honey-dew. They will stop in one place, the same as gnats or mosquitoes, which you have often seen about as high as a man's head.

Now, if any person really wants to test the correctness of this, let him go to a willow grove and he will find those insects (or willow lice) just before sun-down; and getting the willows between him and the sun, he will see them rising from every part of the tree, in small squads, and collecting till they form a large swarm. Then they will be seen discharging continually a fluid which resembles a fine sprinkle of rain. I have often seen those same insects discharging a fluid on a limb, where they were hatching; and then saw large ants, wasps, and yellow jackets working on it. And I often wondered how it got on the very tops of the trees, where no insects were to be found. I think this observation will settle the matter about the origin of honey-dew.

Bees have done very poorly here until now. The golden rod is in full bloom, and the bees are doing well.

H. FAUL.

Council Bluffs, Sept. 6, 1870.



[For the American Bee Journal.]

Caution.

MR. EDITOR:—Through the columns of your indispensable Journal, allow me to say to my brother bee-keepers, and “all whom it may concern,” have nothing to do with a hive called the “Multilocular Protoplasmic Protean Hive,” though it is no doubt superior to any or all you have in use. Let us not step upward only one step at a time to the use of this hitherto excelsior hive; but let us take at least two steps at once, to that hive and those new principles that “beat” *all*. Yes, all the long and toilsome labor of a Huber and a Dzierzon is totally eclipsed; and entirely snuffed out are such lights as Langstroth, Gallup, Quinby, Wagner, and many others, who formerly shone so brightly as “instructors.” Your theories, gentlemen, are forever “cast in endless shade.” The great revolution of nature that moves all things, has thrown before my vision this wonderful apistical domicile. I have scanned it closely, and now let me say to you, Rev. L. L. Langstroth, talk no more of laterally movable frames, since this great hive has “a place for every frame, and every frame in its place.” And you, “far-famed Gallup,” say no more of division boards and economy of heat. ’Tis useless, as these frames are made extra large, and small frames for surplus set in the top of the large ones, which space is left in free communication with the brooding apartment, till again filled with surplus. Speak not, Mr. Wagner, of compactness of form, as this marvellous habitation stands erect, human like. And now the sturdy German (Dzierzon) must yield the palm and transfer it over into Indianapolis, (Ind.) the centre of bee-gravity—the place where one hundred colonies are made from one in a single season! Can we not plainly see the dawning of a day when “the land shall flow with honey,” and each and every individual will supply himself freely with this “sweetest of all sweets,” and the apiarian turn his attention elsewhere for a livelihood?

JAMES HEDDON.

Dowagiac, Michigan.



[For the American Bee Journal.]

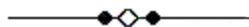
Correction Requested.

MR. EDITOR:—My attention has been called to an alleged error of statement in my article on page 72, Vol. VI., of the Bee Journal, wherein I say, “Mr. Langstroth was among the first to introduce to the notice of the bee-keepers of America the invaluable honey extractor.” Now I claim that the statement is strictly true. Mr. Langstroth was *among* the first to introduce the honey extractor to the notice of the bee-keepers of this country, taking upon himself the responsibility of manufacturing from a bare description, and extensively advertising the machines for sale; thus risking pecuniary loss in case it should prove unpopular, before any other person in this country, except the editor of the American Bee Journal, spent a single dollar upon them.

Still, in order to give every man due credit for any assistance given to bee culture, I will here, with pleasure, state a fact in this connection that had escaped my recollection at time of writing the previous article, namely, that the first *mention* of the machine of Von Hruschka in the English language was made in the American Bee Gazette,⁴ page 85, September No., 1866, edited by Rev. E. Van Slyke, in an article translated from the German, by the editor. And to this article, Mr. Langstroth was most probably indebted for his first idea of the honey extractor, as Mr. Van Slyke writes me as follows—“Mr. Langstroth himself, who visited me at my office the very next month after the publication, spoke in terms of the highest enthusiasm of the article, and said that from my description as published he was about to construct a machine for honey extraction.” &c.

R. BICKFORD.

Seneca Falls, N. Y., Oct. 5, 1870.



[For the American Bee Journal.]

Why are Two Queens Sometimes Found in One Hive?

MR. EDITOR:—Mr. A. Green, in the October number of the Journal, gives an account of finding two queens in one hive. Other correspondents have also given us their knowledge of similar facts; but none have, I think, given us any reasons for such exceptions.

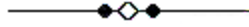
Last fall I bought an Italian queen from a reliable breeder. She came recommended as A No. 1. I received her on the 8th of September. All the workers sent with her were dead, except two; and she was herself so benumbed by cold that I had quite a time of it bringing her back to vitality. Finally I succeeded in getting her quite lively, and introduced her to a tolerably weak swarm. On the 10th of October finely marked Italians were flying in front of the hive. I spared no pains in wintering. (I winter out-of-doors.) In April she had filled three cards of brood. I then gave her a card of drone-comb. She would not look at it, and I moved it back and put in its place a card of worker-comb, which she filled with eggs almost instanter. I then put the drone-comb in the middle of the cluster, and got about fifty drones. Of course I was stimulating, and kept plenty of honey in the hive. I put in other worker-comb, but she refused to lay any more. I then took out a frame to start a nucleus, and in about a week after, when examining the old stock, I found queen cells started and the old queen on the comb, apparently all right. In due course a young queen was hatched, and after destroying the queen-cells, she remained with the old queen ten days before she was fertilized, and at least a week after she was laying. At the end of three weeks the old queen was gone.

Now, what does this prove? Simply that the queen was chilled in coming by mail, which interfered with her prolificness, rendering her supersedure a necessity for the future welfare of the colony. She was tolerated in the hive by the new queen and bees, having lost that distinct individuality peculiar to the queen bee, and consequently become to them (the workers and young

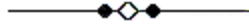
queen) no more than a common bee. I cannot help but conclude that when such exceptions occur, the course relatively is the same.

FREDERICK CRATHORNE.

Bethlehem, Iowa, Oct. 9, 1870.



It cannot be too deeply impressed on the mind of the bee-keeper, that a small colony should be confined to a small space, if we wish the bees to work with the greatest energy, and offer the stoutest resistance to their numerous enemies. Bees do most unquestionably “abhor a vacuum,” if it is one which they can neither fill, warm, nor defend. Let the prudent bee-master keep his stocks strong, and they will do more to defend themselves against all intruders, than he can possibly do for them, even though he spend his whole time in watching and assisting them.—*Langstroth.*



[For the American Bee Journal.]

The Coming Convention.

Mr. EDITOR:—We would like to attend the prospective convention of bee-keepers, which is to assemble the coming fall or winter, and to take by the hand some of the many correspondents we have followed through the columns of your Journal, and hear their opinions by the word of mouth, but we must forego that pleasure at present. We are poor and have not straightened up yet the ravages of war. We are rebuilding as fast as our means will admit, and hope in a few years more to see our once desert looking country “blossom as the rose.” We have lost our substance, the toil of years, and in bee parlance, though driven out and robbed of comb and honey, are allowed to return in a bad season, to recuperate.

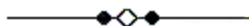
When these bee conventions become yearly in our country, (and I hope they will,) we will be sure to attend, if within the range of our flight. We would be delighted to see the different specimens of honey and bees which should be in attendance, and ahead of anything to see except the phiz of Novice, Gallup, Grimm, and their ilk, side by side the different hives in working order. A great majority of the hives with movable frames are patented, many are not, and we would like to see them on exhibition, opened, and the points of excellence each contains, shown. We don't mean the sub-venders of different patents, who are travelling over the country, and attend at the different fall fairs, who never kept or owned a hive of bees, know nothing of the nature and habit of the insects, and who move up to you and talk as learnedly on the bee as Langstroth or Dzierzon could; but men of experience and veracity, who have tried and used for several seasons the hive on exhibition, through poor as well as rich harvests; and hives of different forms and capacity, which you could criticise, and the good qualities, or the real or imaginary defects of which a man might point out, without danger of being called a mutton-head and ignoramus. There are several different patents in our country, and if they are not thrown over the fence the first season, they are sure to go the way of all trash the second. Some unfortunate purchasers try to get their money back by transforming the hives into troughs to feed the cow in; others convert them into boxes for

hen-nests. In many of these cases, however, it is through the ignorance of the keepers that they do not succeed.

One year ago, Esq. Boring, a Justice of the Peace from one of our rural districts, thought to outstrip his neighbors in honey and bee-keeping, and ordered a hive with which you could control swarming, catch the drones, keep out moths, and the Lord only knows what its owner didn't claim for it. Draw out the chamber, take out honey enough for supper, and replace the drawer, and all is right, nice, and snug! I believe they call it the Buck-eye, patented by Mitchell. Esq. Boring was eager to have bees in, and couldn't wait for a natural swarm, but drove in a fine stock. He was so well pleased with it and its workings, that he Buck-eyed his whole apiary; and upon inquiry a few days since, he informed me that he would lose nearly all his bees. The first time he drew out the chamber everything worked fine. The second time it was rather tight and glued up. A month after that he thought it would take a small yoke of steers to pull out the chamber of frames, and during the summer nearly the whole fell a prey to the moth-miller. However, he should not condemn the hive after this slight trial. It has been an unusually poor season, and none but the strongest stocks stored any surplus.

W. P. HENDERSON.

Murfreesboro, Penna., Oct. 6, 1870.



[For the American Bee Journal.]

The Queen Nursery.

Under the above heading, Mr. Gallup, in the Journal for October, gives his experience with the queen nursery, which, with him, appears to be a perfect success. I wish to give my experience, and ask Mr. Gallup and others why it is so different from his.

I made fifty cages $1\frac{1}{2} \times 1\frac{3}{4} \times 1\frac{3}{4}$ inches, four sides of very thin wood, and one side covered with wire gauze, and the other with a piece of glass slipped in grooves in the two wooden sides, so as to be moved up or down for a door. In each of these cages I placed a piece of honey in comb (unsealed), with the cells in natural position; and then placed the cages in frames, on slots inserted across them, so as to hold three tiers of six each, or eighteen to a frame. I then took out two centre frames from good, strong hives, and put one of these frames containing cages in their place. Some very strong colonies, some were medium. To some I gave upward ventilation, by leaving off the honey boxes and raising the cap. On others I left the honey boxes. I then awaited the result. Some queens hatched in fourteen days from starting the cell; some in sixteen days; two or three in twenty-four days; and some *never* hatched.

Many of the young queens died in the cages in from twelve to twenty-four hours after hatching; very few lived to be five days old—the time given by many writers for them to mate with the drones; only six or seven out of about one hundred lived two weeks. The queens, when first hatched, were put in fertilizing cages such as described by N. C. Mitchell, but *never* were fertilized.

Now Mr. Editor, will Mr. Gallup or some one else tell me why my experience differs so widely from that of Mr. G.?

Sister cells, cut from the same comb as some of those that were put in the cages, hatched in from fourteen to sixteen days, were duly fertilized, and are now alive and well. Hence it could not be any defect in the stocks they

were raised from. In some of the cages, I put two or three workers, to feed the young queens; but still the latter would die, and leave the workers to eat the honey left in the cages.

If queens require any other food than honey, why did not the bees give it to them through the wire gauze on which they clustered in great numbers? Some of the cages were put in colonies that had fertile queens at liberty, but most of them were put in queenless hives.

The cells were mostly put in on the ninth day from starting the cell.

I shall be pleased to see replies to this in the next number of *AMERICAN BEE JOURNAL*.

H. NESBIT.

Cynthiana, Ky.

[For The American Bee Journal.]

Do the Right!

Friend Bickford, I wish to shake your honest fist!

Your matter is *sound*, your argument JUST!!

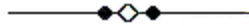
To render substantial aid to our “venerable Tutor” is an imperative duty. Let us see to it then, *at once*, and

DO THE RIGHT!

I don’t feel at liberty to enlarge on the subject, being “only an Englishman.”

WALTER HEWSON.

Wickham-breaux, Kent, England, Sept. 28, 1870.



[For the American Bee Journal.]

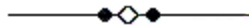
The African Honey Tree.—Inquiry.

In the "*Poultry Bulletin*," J. M. Wade, of Philadelphia, writes—"A man, I can hardly say *gentleman*, came into the store yesterday, with seventy-one humming birds, which he had shot the day before in his own yard. He said some years ago he brought a honey tree from Africa, and thousands of humming birds would come to it in one day. Where did so many come from?"

As it may be in the interest of bee culture to know what can be learned about the honey tree of Africa, will some one who is informed give the readers of the *AMERICAN BEE JOURNAL* his knowledge of it? stating its growth, whether bees visit it, its uses, whether it is hardy, length of time in flower, in what month and at what age it blooms, and how it is propagated?

E. PARMLY.

New York.



Early in October, I examine carefully all my hives, to see that they are in suitable condition for wintering. If any need feeding, they are fed at this time. If any have too much vacant room, I partition off that part of the hive which they do not need. I always expect to find some brood in every healthy hive at this time, and if in any I find none, and ascertain that it is queenless, I either at once break it up, or if it is strong in numbers, supply it with a queen, by adding to it some feebler stock. If bees, however, are properly attended to, at the season when their young queens are impregnated, a queenless colony will seldom be found in the fall.

LANGSTROTH.

THE AMERICAN BEE JOURNAL.

Washington, Nov., 1870.

The residence of the Rev. Mr. Semlitsch is not at Gratz, in Styria, as, in consequence of a slight omission, was erroneously stated in our last issue; but at Strasgang *near* Gratz.

The attention of those who are unfortunately suffering from the prevalence of foul-brood in their apiaries, will doubtless be arrested by the communication, in this number of the Journal, from Dr. ABBE, of New Bedford (Mass.), announcing that he has succeeded in curing that disease, as it existed in several of his colonies; and that an efficient and easily applicable remedy has at length been devised for the dreaded and devastating evil. Dr. ABBE deserves the cordial thanks of bee-keepers, both in this country and abroad, for so generously and promptly making known his remedy and the mode of administering it.

Last fall we suggested to those who found it necessary to supply their bees with winter food to add a portion of glycerine to sugar syrup or dissolved candy, to prevent crystallization; and we learn that it was advantageously used. We have since learned that gum tragacanth is now employed for the same purpose, by some of the German bee-keepers. This gum, dissolved in water, forms a thick mucilage, which may not mingle so readily with the food as glycerine does; and the latter is hence a more manageable and probably cheaper article, especially as it forms besides an excellent spring stimulant, though still too high-priced to be freely used.

A bee-keeping friend has procured for us a quantity of seed of the Partridge Pea (*Cassia chamæcrysta*) mentioned by one of our western correspondents, (Mr. Ingels, of Oskaloosa, Iowa,) as an excellent honey plant. It was in bloom here from the middle of July to the middle of October, and frequented by that bees, in crowds, all the time.

This plant is usually classed among weeds, and where it occurs, is regarded by some as one of the *pests* of the farm; but as it is an annual, it ought not to be difficult to get rid of it by proper management, when its presence is undesirable. Blooming during the interval between spring and fall pasturage, it constitutes an important resource for bees, here and in other districts, at a period when the native vegetation fails to furnish supplies.

In the third volume of the Transactions of the American Philosophical Society, Dr. Greenfield of Virginia speaks of the Partridge Pea as furnishing means to recruit worn-out lands, by its decomposition in the soil when plowed under. It was, we understand, originally introduced for that purpose, in the District of Columbia, by the Hon. Benjamin Stoddert, while Secretary of the Navy; and it would probably answer well as a substitute for red clover, where from poverty of soil, the latter could not yet be grown.

We hope to be able to make satisfactory arrangements for the distribution of the seed among bee-keepers desiring to make trial of the plant, and if successful, will state particulars in our next.

We learn from Mr. Adam Grimm, of Jefferson (Wis.,) that his crop of surplus honey, this year, is over 15,000 lbs., and that he “could take at least 10,000 lbs. more from his hives, and still leave the stocks heavy enough to winter well.” Such a result as this must be calculated to unsettle the notions of those who “have kept bees many years, and *know* there is nothing to be made by it!”

We intended to give a brief history of the opposition to the meeting of the National Convention of Bee-keepers at Indianapolis, showing when and where it originated, and what were the obvious motives and objects of those most active in the business. But as it appears to be a “fixed fact” now that the Convention will be held at the time and place designated, we shall save ourselves the trouble of hunting up musty records in the limbs of things forgotten.



Since the above was put in type we have learned incidentally that it was resolved at Utica by the N. E. Bee-keepers Association to hold another Convention elsewhere, though particulars have not reached us. We sincerely regret this proceeding on various accounts.



CORRESPONDENCE OF THE BEE JOURNAL.

TRENTON, ILLS., Sept. 12, 1870.—The forepart of this season I think was the poorest I ever saw in this neighborhood. Last winter was a very warm and open one, and the bees dwindled down very much, so that nearly all stocks were quite weak before spring. Then we had a severe snow storm on the 17th of April, with two or three freezing nights, that killed nearly all the peach blossoms; and this was followed by a period of cold high winds through May. The first two weeks of June there was cloudy, drizzling, chilly weather, so that bees could not fly more than about half the time. The consequence of all this was, late swarms and very few of them. Not more than one-sixth of the stocks swarmed, and many of the latest of them starved. It was very dry from the middle of June to the 13th of August. Then, for a week, it rained nearly every day; at the end of which some of my hives had not more than a pound of honey remaining. Since that time they have been doing very well. Most of my hives were filled up, so that they commenced working in the surplus boxes about the middle of last week, and some of them have now as much as fifteen pounds in the boxes.

I would like NOVICE to tell us how he gets his board and frame into the top of his hive, if his hives are all of one size. I have a few of the two-story hives made by the National Bee-hive Company at St. Charles, Illinois, and I cannot get a frame into the top story in any other way than perpendicular, as the top bar of the frame is longer than the inside of the hive. I have tried one to see how it would work.—C. T. SMITH.

DOWAGIAC, MICH., Sept. 12—We have had just half a surplus honey harvest, here, this season. Since I have been in the bee business, I have learned that the surplus harvest depends entirely upon the clover and basswood blossoms, in this vicinity; which is probably the case all over the State. When we have a wet season clover fails, but basswood produces well; and when a dry season, *vice versa*. Reverses from abundance to starvation take place within a few miles of each other. I am located now in the midst of clover and basswood, together with the best spring and fall pasturage I have

ever seen. After losing seven-eighths of my bees last winter, you can easily guess the condition of the remaining six colonies. Four of them were merely skeletons, and the other two very inferior stocks. Yet, with the aid of a three cent feeder of my own invention, (which works to perfection,) and one and a fourth dollar's worth of sugar, I have succeeded in marketing five hundred and twenty-three (523) pounds of box honey; and with the aid of old combs have increased my stock to twenty-two (22) colonies, all strong and heavy—too heavy I fear, for their own good; and I have as yet no emptying machine. This, I think, is doing very well (see Langstroth's "HIVE AND HONEY BEE," page 177) for a bee-keeper of only two years' experience.—I came near forgetting to mention that I have Italianized all my new stocks. I use top-bar hives mostly. Am using four or five frame hives on the sly!—J. HEDDON.

WINCHESTER, IOWA, Sept. 13.—The season of 1870 has not been any of the best here, nor of the poorest either, as swarming and honey gathering has been moderately good. The American Bee Journal well deserves the support of bee-keepers.—I. N. WALTER.

ROCHELLE, ILLS., Sept. 17.—This has been the poorest season that we have had here for some years. I got only five new swarms from forty stands, and merely one hundred pounds of honey. Since the buckwheat came into blossom the bees have done well. They will average about fifty pounds to the stand; and that is doing very well, in such a year as this has been. Alsike clover is now in blossom, and the bees are working very busily on it.—R. MILLER.

BREESPORT, N. Y., Sept. 20.—My bees have done well in gathering honey, this season; but gave me no swarms during swarming time.—J. H. HADSELL.

OSKALOOSA, IOWA, Sept. 28.—I have one hundred and ninety colonies of bees that have done well this year, and are in fine condition for winter. I stored away one hundred and twenty-nine colonies in my cellar last fall, and the same number came out in good order in the spring. I sold them off to about one hundred, from which I came on to winter with the above number (190), principally Italians.

Enclosed please find specimen of a bee plant. What is it? It blooms from first of July to last of August profusely and is visited by bees thrice as much as buckwheat. I have tried borage, melilot, alsike, mustard, and find nothing to equal it. I calculate to cultivate it, in order to give it a fair and full trial. I have secured about a peck of seed. The great advantage is that it blooms at a time when most needed in this country. I grew it this year alongside of buckwheat that bloomed at the same time.—S. INGELS.

[The plant enclosed is the *Cassia chamæcrista* or Partridge Pea. It is an annual, growing in most sandy soil, and is common in the south. It grows here on the eastern branch of the Potomac (the Anacostia), and bees derive plentiful supplies of forage from it during eight or ten weeks in summer, and it is then almost their only resource. They gather pollen from the blossoms, but the honey is secreted by a small cupshaped gland situated below the lowest pair of leaflets, and is supplied abundantly for a long period.—Some of the farmer's hereabouts affect to consider it a pernicious and ineradicable weed; but as it is an annual and known to be an excellent fertilizer when plowed under, it would seem to indicate slovenly management not to be able to subdue it readily where not wanted.—ED.]

VERVILLA, TENN., Sept. 24.—I consider the Journal cheap at any price for the bee-keeper, and wish it could be published oftener.—DR. J. M. BELL.

WARSAW, MINN., Oct. 3.—This has been a poor season for bees here, except in basswood time.—L. B. ALDRICH.

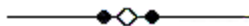
CEDARVILLE, ILLS., Oct. 5.—My bees have done well this season.—ROBERT JONES.

MEREDITH, PA., Oct. 4.—Bees did very well on white clover in this section this season, but very poorly on buckwheat. My sixty stocks did not give me sixty pounds of buckwheat honey surplus, all told; although they are all in good condition for wintering.

I do not think that alsike clover has been over-estimated for bee pasturage. I had three-quarters of an acre of it this season, and I never saw a piece of land so covered with bees as that was while it was in bloom, and they gathered honey from it very fast.—M. WILSON.

ORCHARD, IOWA, Oct. 6.—It is raining heavily to-day, yet the weather is warm and we have not had a particle of frost yet. Bees have done storing surplus honey for the season.—I shall give the result of the season's operations as soon as I can get the time. At present I am up 4 A. M., and do not get home till 8 and sometimes 9 o'clock P. M. I must have a little relaxation from such excessive hard labor, before I can confine or control my thoughts sufficiently to write for publication. From the past season's operations with the honey extractor, I can endorse all that Novice claims over and above the old mode of getting surplus in the comb—E. GALLUP.

NEW BEDFORD, MASS., Oct. 6.—The season for bees has been remarkable. Commencing well, the dry weather soon made forage very scarce during the blooming of clover and basswood, so that by the first of September there was little or no surplus stored, and all the colonies were very light. But during that month, mostly after the fifteenth, the bees gathered honey as fast or faster than they ever do in this locality in June. It was obtained from the wild aster; and the stocks are now heavy and in fine condition for winter. Even now there seems to be no cessation of their labors. This is true of all the neighboring towns; nearly every hive in them having been examined by me during my professional drives.—E. P. ABBE.



[For The American Bee Journal.]

How May Progress be Taught?

MR. EDITOR:—As the columns of the Bee Journal are made the medium of disseminating apicultural knowledge, by asking and answering questions, I have this question to ask in reference to the class of bee-keepers who use box and gum exclusively. How shall we reach these, and dispense the necessary knowledge among them? Let us endeavor to devise some effective means. Your Journal is doing the work as far as they can be induced to take and study it; but the number is comparatively limited. Many of these people, when they see an improved bee-hive, unconsciously exclaim to the owner, who happens to be a practical bee-keeper:

“Mr. B.—What do you call that?”

B. “That, sir, is a bee-hive.”

Q. “What do you have so many sticks in it for?”

B. “Those are what we call *frames* for the bees to build their combs on; each frame separately giving them the means by which the combs may be removed from the hive, for the purpose of making artificial swarms, furnishing honey from the rich to the poor colonies and strengthening weak ones.”

Here the querist exclaims in perfect amazement: “What will the bees be doing while you are lifting their combs out?”

B. “If you treat the bees right they will not harm you; besides we can have a protection, made of wire cloth, or what is more handy, a piece of bobbinet to place over the face; and by keeping the hands wet, the bees will not sting, unless they are badly treated.”

Q. “What a fool I have been. I have kept bees all my life, and never before knew what I needed. I suppose if you can lift out the combs, as you say you can, you could find the king’s house and perhaps the king himself?”

B. "There is no such bee in the hive."

Q. "What! no king bee! Why I always understood that a colony of bees without a king and ruler, whose mandates are strictly obeyed, will not be worth anything."

B. "The bee you allude to is the mother of the colony and is called the queen; but she has no house or particular spot in the hive in which she dwells. The worker-bees, however, construct what are called queen-cells, in which queens are reared; but they never remain in them, except only while in embryo."

Q. "Why, Mr. B., you seem to know as much about bees as the man I heard a neighbor speak of. He said there was a man living in Iowa that reared king bees (perhaps you would call them queen bees) of a superior and different kind from the common bee, and brought from some other country."

B. "Yes, we rear our own queens, or in other words we cause the bees to do so, by our artificial process. This we do for the purpose of furnishing fertilized queens to old stocks, when their queens are taken away, as is the case in producing artificial swarms."

Q. "Then you can make bees swarm, and rear queens at your will?"

B. "Yes."

Q. "But do you never find a hive that is not in the notion of swarming? I always thought that bees knew when they wanted to swarm, better than man did."

B. "Bees have only instinct, and were not intended in the beginning to produce their own swarms. They were created for the benefit of man, and if that had been the way swarms were intended to be made, they would be made in conformity with natural laws that govern them, and swarming would always be successfully performed in perfection. Man was given knowledge, by means of which it was intended he should manage his bees in his own way, independent of any will they may have. The penalty for man's neglect in this respect is the loss of his bees in various ways—such as

swarming and departing to parts unknown, loss of queen, extermination by robbing, &c. Man, therefore, endowed with knowledge and judgment, knows more of the management, for his benefit, of the internal parts of the hive, than the bees, with mere instinct, can possibly know.”

Q. “I perceive, sir, that these are the days of our ignorance spoken of in Holy Writ, though I was never able to see it till now. Some of my neighbors, a few years ago, purchased bees which were in common boxes and gums. They brought them home and set them down in a remote corner of the yard or garden, to live or die, as they might or could, with no attention whatever, except when the time came to secure some of their delicious stores, which, with shame I confess, is the practice in all the neighborhood now.”

B. “Your statement is only too true, if indeed the facts are not worse.”

This is a fair specimen of the questions asked by common bee-keepers.

While the inventive genius of the age has given power to water in the form of steam, causing the face of the earth to be alive with machinery and wheels that are almost daily circumscribing its surface at lightning speed—yea, the lightning itself has, as it were, been snatched from the heavens and made to do the bidding of man—yet the bee-hive, till within the last fifteen years, has in a measure remained as it may have been in the garden of Eden. The invention of the frames was the dawn of a new era in bee-keeping, by means of which we have advanced step by step up the hill of science to the present advanced stage, while progression still looms up and fades away in the distance. The mysteries of the hive that remained hidden from the beginning till now, are, many of them, solved and being solved, and all the various causes of the destruction of colonies plainly disclosed. The practical man, properly informing himself, need not lose a hive; while, in the old way, twenty-five per cent, of all the bees kept in the country are lost every year. While we have reached these advances, there are many things yet in embryo, that will be reached by and by—such as the control of fertilization, which enables the bee-keeper to select both queens and drones, and secure the purity of the race we prefer to cultivate. We also expect a forcing-box, hiver, and swarmer, all combined; and means which will enable the bee-

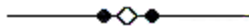
keeper to compel a plurality of queens in every colony, without division, in the same apartment.

But I am wandering from my purpose, which was simply to start the inquiry—how shall we reach, and dispense the necessary knowledge among those who still keep their bees in unimproved hives? The State governments should foster bee culture as they foster other agricultural pursuits. Why not have a separate department for bee culture in every State, under the charge of a man qualified to superintend it and diffuse its advantages in the community? In some of the German States the number of hives will average hundreds to the square mile, and that too in soil comparatively sterile. How was this brought about? Simply by encouraging and fostering the business. And cannot the American States produce the same results? Millions of barrels of honey go to waste annually in this country, merely from the want of bees to gather the nectar of flowers. What, say you, bee-keepers of Iowa, shall we not make a united effort to secure the means by which those who have bees in our beautiful State shall be furnished with power (knowledge) to effect the gratifying change? The bees of every hive now in the State, producing ordinarily ten, twenty, or thirty pounds, may be made to produce annually from one hundred to two hundred pounds.

Mr. Gallup will please accept our thanks for his practical and instructive communications in the Journal. Will he not favor us with an article on this subject. Let Iowa be the first to take a stand in favor of promoting bee culture.

J. W. SEAY.

Monroe, Iowa.



[For the American Bee Journal.]

Argo's Puzzle.

R. M. Argo has found a job for Gallup.

That bees will sometimes build worker-comb when there is no queen present is a positive fact, but the rule is almost invariably drone comb. The fact that they built one-third drone comb is no proof that they did not have an old queen. If they are gathering honey abundantly, they are very apt to build too much drone comb; and sometimes they do so in such cases, even with a young prolific queen. But with such a queen, when they are gathering just sufficient to build comb and store but little honey, the rule is almost invariably worker comb exclusively.

That bees will frequently make preparations for swarming immediately after being hived is another positive fact, especially when the season is good and the newly hived swarm is large. The first case of the kind that came under my observation, occurred a number of years ago in Canada. I hived an extra large swarm for a neighbor, sometime in the forenoon. About four o'clock in the afternoon the shout came across the mill stream, "my bees are going off!" I left all, and followed them to a large pine stub. I cut down the stub, split it open, took out the bees, put them in the same hive. That night they were sold as an *unlucky* swarm, removed $3\frac{1}{2}$ miles, and in just eight days from the time they were replaced in the hive, they sent out a large swarm, which left for the woods. The bees then belonged to my cousin. They left on Saturday. On Sunday I went to church close by my cousin's, and he informed me that his bees had filled their hive and swarmed, and the swarm left for parts unknown. I was rather incredulous, but after church went and made an examination. Sure enough, the hive was completely filled and several sealed queen cells were in sight, with several more unsealed near the bottom of the comb. The hive was a box twelve inches square by fourteen inches high, and when the swarm was hived I had to put on a large box before the bees could all be got in the hive. That box was nearly filled with comb, but the bees that went off took the honey with them. On the fifteenth day they sent out a second swarm. So much for

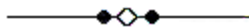
purchasing an *unlucky* swarm!—Since then I have had several cases of the same kind come under my observation; one in the summer of 1868, and another this summer. The one in 1868 was not a large swarm, and they did not fill their hive before sending out a swarm. The case this season was a large artificial swarm made by putting together bees from several hives, with a queen.—I should be strongly inclined to think that, in your case, they started queen cells for the purpose of superseding the old queen. When a queen has begun to fail at about swarming time, and forage is abundant, they cast a swarm. In my case, in 1868, it was no doubt caused by the bees superseding the old queen. I had a case this season, where the first swarm came out with a young queen, leaving the old queen in the hive, with plenty of sealed queen cells. In another case, when making an artificial swarm, I found the old queen and a young one both, fertile, with several sealed queen cells.

E. GALLUP.

Orchard, Iowa.



The amputation of *one* of the antenna of a queen bee appears not to affect her perceptibly, but cutting off both these organs produces a very striking derangement of her proceedings. She seems in a species of delirium, and deprived of all her instincts; everything is done at random; yet the respect and homage of her workers, towards her, though they are received by her with indifference, continue undiminished. If another in the same condition be put in the hive, the bees do not appear to discover the difference, and treat them both alike; but if a perfect one be introduced, even though fertile, they seize her, and keep her in confinement, and treat her very unhandsomely. “*One may conjecture from this circumstance, that it is by those wonderful organs, the antennæ, that the bees know their own queen.*”



That which is profitable only to the speculating business, though it be theoretically plausible, deserves not to be recommended or accepted, if it be not calculated to produce beneficial results to the practical bee-keeper.

FOOTNOTES:

- [1](#) My method and the use of Dr. Davis' QUEEN NURSERY.
- [2](#) Oranges, bananas, pineapples, and other tropical fruits are forced in hot-houses; but they never reach the size, flavor, or perfection of nature.
- [3](#) In a subsequent communication in Vol. V., No. 10, Mr. B. says that in place of old woollen garments, he covered the frames last winter "with a sort of cotton batting comforters made precisely like a comforter for a bed; and that he likes these much better than old carpeting or old clothes." He had one made for each hive, costing about twenty cents a piece. "By lifting one corner of these comforters, the condition of the hive can be seen at a glance. The bees are always found clustered up against these warm comforters, and communicate over the tops of the frames, instead of through the winter passages."
- [4](#) Shortly thereafter merged in the American Bee Journal.

Transcriber's Note:

Obvious printer errors corrected silently.

Inconsistent spelling and hyphenation are as in the original.

*** END OF THE PROJECT GUTENBERG EBOOK THE AMERICAN
BEE JOURNAL, VOL. VI., NUMBER 5, NOVEMBER 1870 ***

Updated editions will replace the previous one—the old editions will be renamed.

Creating the works from print editions not protected by U.S. copyright law means that no one owns a United States copyright in these works, so the Foundation (and you!) can copy and distribute it in the United States without permission and without paying copyright royalties. Special rules, set forth in the General Terms of Use part of this license, apply to copying and distributing Project Gutenberg™ electronic works to protect the PROJECT GUTENBERG™ concept and trademark. Project Gutenberg is a registered trademark, and may not be used if you charge for an eBook, except by following the terms of the trademark license, including paying royalties for use of the Project Gutenberg trademark. If you do not charge anything for copies of this eBook, complying with the trademark license is very easy. You may use this eBook for nearly any purpose such as creation of derivative works, reports, performances and research. Project Gutenberg eBooks may be modified and printed and given away—you may do practically ANYTHING in the United States with eBooks not protected by U.S. copyright law. Redistribution is subject to the trademark license, especially commercial redistribution.

START: FULL LICENSE

THE FULL PROJECT GUTENBERG™ LICENSE

PLEASE READ THIS BEFORE YOU DISTRIBUTE OR USE THIS WORK

To protect the Project Gutenberg™ mission of promoting the free distribution of electronic works, by using or distributing this work (or any other work associated in any way with the phrase “Project Gutenberg”), you agree to comply with all the terms of the Full Project Gutenberg License available with this file or online at www.gutenberg.org/license.

Section 1. General Terms of Use and Redistributing Project Gutenberg electronic works

1.A. By reading or using any part of this Project Gutenberg electronic work, you indicate that you have read, understand, agree to and accept all the terms of this license and intellectual property (trademark/copyright) agreement. If you do not agree to abide by all the terms of this agreement, you must cease using and return or destroy all copies of Project Gutenberg electronic works in your possession. If you paid a fee for obtaining a copy of or access to a Project Gutenberg electronic work and you do not agree to be bound by the terms of this agreement, you may obtain a refund from the person or entity to whom you paid the fee as set forth in paragraph 1.E.8.

1.B. “Project Gutenberg” is a registered trademark. It may only be used on or associated in any way with an electronic work by people who agree to be bound by the terms of this agreement. There are a few things that you can do with most Project Gutenberg electronic works even without complying with the full terms of this agreement. See paragraph 1.C below. There are a lot of things you can do with Project Gutenberg electronic works if you follow the terms of this agreement and help preserve free future access to Project Gutenberg electronic works. See paragraph 1.E below.

1.C. The Project Gutenberg Literary Archive Foundation (“the Foundation” or PGLAF), owns a compilation copyright in the collection of Project Gutenberg electronic works. Nearly all the individual works in the collection are in the public domain in the United States. If an individual work is unprotected by copyright law in the United States and you are

located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work. The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg™ License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official version posted on the official Project Gutenberg website (www.gutenberg.org), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, "Information about donations to the Project Gutenberg Literary Archive Foundation."
- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.
- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.
- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project

Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to

you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a) distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

Section 2. Information about the Mission of Project Gutenberg

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the

efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at www.gutenberg.org.

Section 3. Information about the Project Gutenberg Literary Archive Foundation

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact links and up to date contact information can be found at the Foundation's website and official page at www.gutenberg.org/contact

Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment

including outdated equipment. Many small donations (\$1 to \$5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit www.gutenberg.org/donate.

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including checks, online payments and credit card donations. To donate, please visit: www.gutenberg.org/donate.

Section 5. General Information About Project Gutenberg electronic works

Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a

copyright notice is included. Thus, we do not necessarily keep eBooks in compliance with any particular paper edition.

Most people start at our website which has the main PG search facility:
www.gutenberg.org.

This website includes information about Project Gutenberg, including how to make donations to the Project Gutenberg Literary Archive Foundation, how to help produce our new eBooks, and how to subscribe to our email newsletter to hear about new eBooks.